Precambrian Research 178 (2010) 119–135

Contents lists available at ScienceDirect

Precambrian Research

journal homepage: www.elsevier.com/locate/precamres

Proterozoic Basic dykes in the Ukrainian Shield: A palaeomagnetic, geochronologic and geochemical study—The accretion of the Ukrainian Shield to Fennoscandia

Sten-Åke Elming[a,*], Leonid Shumlyanskyy[b,c], Svetlana Kravchenko[d], Paul Layer[e], Ulf Söderlund[f]

[a] Department of Applied Chemistry and Geosciences, Luleå University of Technology, SE- 97187 Luleå, Sweden
[b] Institute of Geochemistry, Mineralogy and Ore Formation, 34 Palladina Avenue, Kyiv, Ukraine
[c] Laboratory for Isotope Geology, Swedish Museum of Natural History, Box 50007, 10405 Stockholm, Sweden
[d] Institute of Geophysics, National Academy of Sciences of Ukraine,32 Palldina, Avenue, 03142 Kyiv, Ukraine
[e] Geophysical Institute, University of Alaska Fairbanks, Fairbanks, AK, USA
[f] Department of Geology, GeoBiosphere Science Centre, Lund University, Sölvegatan 12, SE-223 62 Lund, Sweden

ARTICLE INFO

Article history:
Received 15 April 2009
Received in revised form 1 February 2010
Accepted 2 February 2010

Keywords:
Ukrainian Shield
Proterozoic dykes
Plate tectonics
Palaeomagnetism
Geochronology
Geochemistry

ABSTRACT

A palaeomagnetic, geochronologic and geochemical study of basic dykes in the Ukrainian Shield has been undertaken with the purpose of testing a hypothesis for timing of accretion of the Ukrainian Shield with Fennoscandia. On the basis of geological data it has been suggested that the two shields amalgamated at ca. 1900–1800 Ma. With this study of 11 basic dykes the palaeomagnetic database for the Ukrainian Shield is extended. The palaeomagnetic data indicate three generations of dykes. This is supported by the age data and the difference in geochemical compositions. Ages of two ca. 25 m wide dykes (Susly) in the North-Western Block of the shield have been determined, one for which a positive palaeomagnetic field test is demonstrated. The lower of the two $^{40}Ar/^{39}Ar$ hornblende ages of this dyke, 2061 ± 20 Ma, is similar to the U–Pb, zircon age of the granitic host rock (ca. 2067 Ma, U–Pb, zircon), which suggests that the dyke intruded shortly after crystallization of the granite. Combined with the magnetization of the baked contact and that of another dyke a new ca 2060 Ma pole is presented (Plat. = 15.7°, Plon. = 182.9°, A_{95} = 13.7°), which may be regarded as a VGP. An U–Pb age of 1722 ± 12 Ma of baddeleyite from the Khmelnik dyke, which yields a well-defined VGP, is similar to the age of nearby anorthosite complexes. A new 1740–1770 Ma key pole for the Ukranian Shield is also presented (Plat. = 26.5°, Plon. = 169.1°, A_{95} = 3.9°). On the basis of palaeomagnetic and geochronological data a 2060–1720 Ma apparent polar wander for the Ukrainian Shield is defined. This apparent polar wander is significantly different from that of Fennoscandia during the same time interval and indicates that the two shields were not in their present relative positions at ca. 1770 Ma. A scenario is here proposed where the Ukrainian Shield (and Sarmatia) collided with Fennoscandia at ca. 1900–1800 Ma and after that event, at ca. 1720–1660 Ma, it rotated approximately 43° into its present position.

© 2010 Elsevier B.V. All rights reserved.

1. Introduction

For our understanding of the geological evolution of the Earth knowledge of the configuration of supercontinents, how and when they were formed is critical. Geologic evidence suggests that the Archaean cratons are fragments of one (Kenorland; e.g. Williams et al., 1991) or more (Sclavia, Superia and Vaalbara; Bleeker, 2003) supercratons that were formed in the late Archaen to early Palaeoproterozoic and at ca. 1900–1800 Ma, the supercontinents Nuna (Hoffman, 1997), Hudsonland (e.g Meert, 2002; Pesonen et al., 2003) or Columbia (e.g. Rogers, 1996; Rogers and Santos, 2002; Zhao et al., 2002, 2004) are suggested to have formed.

The East European Platform can be divided into three crustal segments (Fig. 1) with autonomous histories of development, Fennoscandia, Volga-Uralia and Sarmatia (Bogdanova et al., 2001) and the timing of accretion of these segments is poorly constrained. In this study, the intension is to improve the basis for a palaeomagnetic test of a hypothesis for the time of accretion of the Ukranian Shield (and Sarmatia) with Fennoscandia as proposed by Elming et al. (1993, 2001). This is done by combining new palaeomagnetic, geochemical and geochronologic data for basic dykes in the Ukrainian Shield and then comparing the calculated position and

* Corresponding author.
E-mail address: Sten-Ake.Elming@ltu.se (S.-Å. Elming).

0301-9268/$ – see front matter © 2010 Elsevier B.V. All rights reserved.
doi:10.1016/j.precamres.2010.02.001

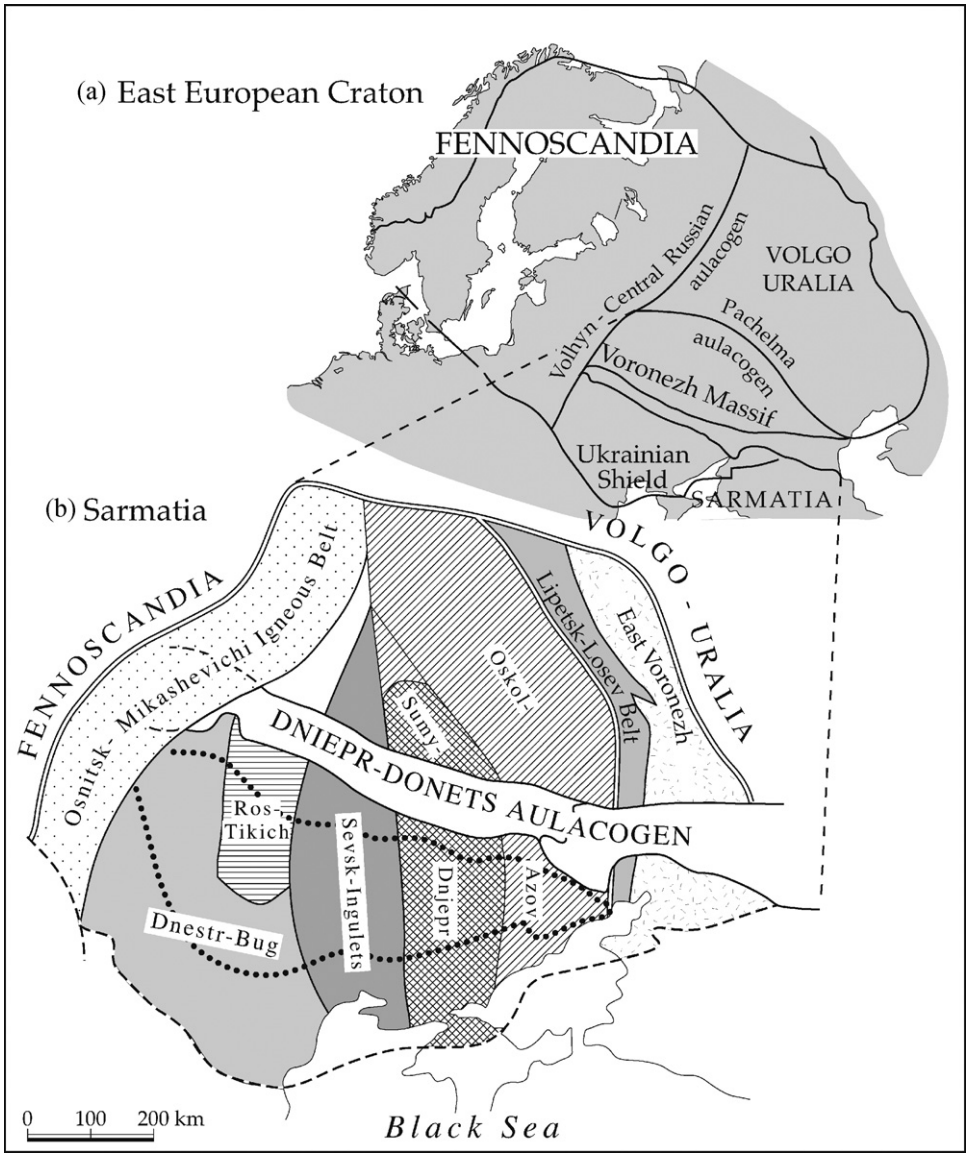


Fig. 1. (a) The East European Platform and the tectonic boundaries separating the Fennoscandian, Volgo-Uralian and Sarmatian cratons (after Gorbatschev and Bogdanova, 1993). (b) The crustal domains in Sarmatia defined on the basis of geochronology, lithology and tectonic pattern (see Shchipansky and Bogdanova, 1996) and the Ukrainain Shield outlined by the dotted line.

orientation of the shield with that of Fennoscandia. The presumed time of accretion in this hypothesis coincides with the formation of the supercontinents Nuna, Hudsonland and Columbia at 1900–1800 Ma ago.

Basic dykes have proven to be excellent palaeomagnetic recorders, which is supported by preliminary results from the Ukraine (Elming and Kravchenko, 2005). Palaeomagnetism, coupled with precise age determinations, is the only method which provides direct estimate of the orientations and ancient latitudes of continents. Here, geochemical analyses of the dykes are also used to provide information on the origin and to put constraints on the interpretation of the ages of the dykes.

1.1. Geology

Autonomous histories of development of the three cratons of the East European Platform (EEP): Fennoscandia, Volga-Uralia and Sarmatia, are indicated by lithological and geochronological differences and by differently trending Palaeoproterozoic structures and boundaries between rock units. The 2100–2000 Ma suture zones, which subdivide the EEP, represent different types of collisional and accretional interactions between the crustal segments (Bogdanova, 1993; Gorbatschev and Bogdanova, 1993).

Sarmatia consists of different Archaean terranes, which are suggested to have been welded together into a coherent unit between 2300 and 2100 Ma ago (Schipansky and Bogdanova, 1996). Sarmatia is cut by the Palaeozoic Dniepr-Donets Aulacogen with a deeply subsided Precambrian basement, separating the Voronezh massif in the north from the Ukrainian Shield in the south. Results from studies of the two basements areas suggests that three of the Archaean terranes have their equivalents on both sides of the rift (Fig. 1), indicating that there has been no or only minor movements between these blocks since the Palaeozoic (Schipansky and Bogdanova, 1996).

The boundary between Sarmatia and Fennoscandia is located beneath the Neoproterozoic Volyn-Orsha depression and is defined by the Osnitsk-Mikashevivichi Belt (OMB; Fig. 1), which belongs to the Sarmatian craton, and trends in a north-eastern direction. Ages of rocks related to the OMB (Claesson et al., 2001) and geophysical data (Stephensson et al., 1996) suggest that this belt represents

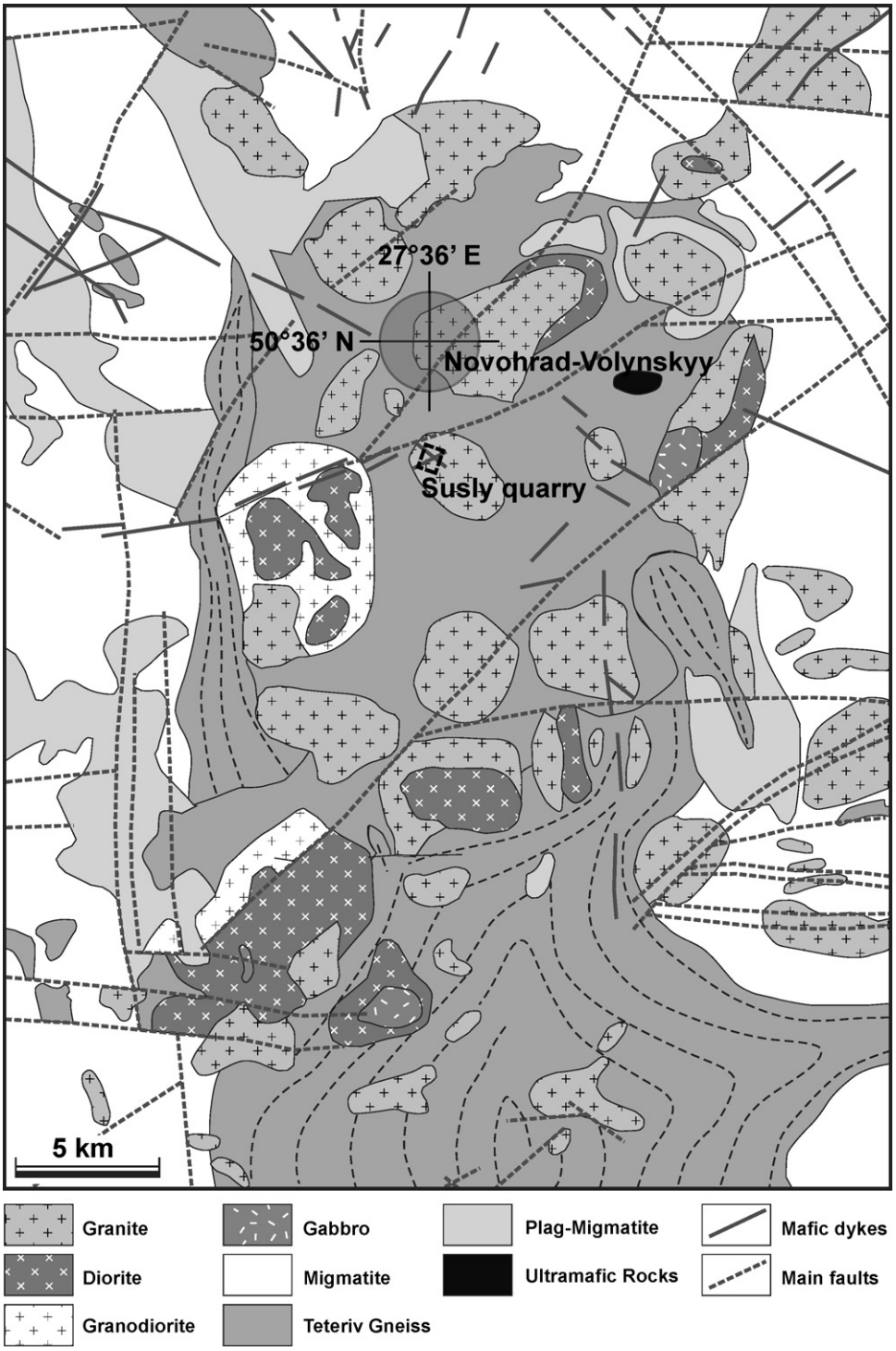


Fig. 2. Geological map of the Novograd-Volynskyy area with the Susly quarry shown by a hatched rectangle. Note the differently trending dykes, two of which appear as cross cutting dykes in the Susly quarry.

a 2.1–2.0 Ga ocean-continent interface. To the Northwest in the Fennoscandian craton, there is a wide region of repeated formation of new crust (2.0–1.9 Ga) in a continental-margin and island-arc setting (Claesson et al., 2001) and of metamorphism (e.g Skridlaite et al., 1997), which indicate an amalgamation of this crust that ended at ca. 1.7 Ga ago.

The Ukrainian shield is located in the central part of Sarmatia and consists of ca. 3500–1700 Ma old metamorphic and magmatic rocks. The shield has been intruded by numerous dyke swarms of various age and composition (e.g. Ahmetshina, 1975; Krutihovskaja et al., 1976). In the late Palaeoproterozoic, anorthosite-rapakivi granite plutons intruded into the central and northwestern parts of the Ukrainian Shield. They have been dated at 1790–1720 Ma (Amelin et al., 1994; Scherbak et al., 1995) and basic dykes are related to them.

The dykes investigated in this study are mainly located in the north-western Volyn and in the Dniestr-Bug domains of the Ukrainian Shield. Chemical analyses of dykes from a previous palaeomagnetic study in the Ingulo-Ingulets and the Dniepr domains (Elming et al., 2001) have also been included.

In the Volyn domain the predominant rock types occurring in the neighbourhood of Novohrad-Volynskyy are supracrustal rocks of the Teteriv Series and granitoids (including migmatites) of the Zhytomyr complex (Fig. 2). Both rock types are cut by a few mafic-ultramafic intrusions and by numerous mafic dykes that tend to form swarms confined to main faults structures. Rocks of the Teteriv Series are represented by mica and amphibole gneisses and micro-gneisses. The age of the Teteriv gneisses was recently determined by Claesson et al. (2006, 2008) to be ca. 2100 Ma, while migmatites were formed at 2080 Ma and Zhytomyr granite intrusions in the Novohrad-Volynskyy area at 2043 Ma. Claesson et al. (2006) also dated a dacitic dyke intruding the gneisses at 2.05 Ga. The Susly granite intrudes into the Teteriv micro-gneisses and reveals sharp contacts against them. The granite itself is cut by several mafic dykes, two of which are included in this study (Fig. 2).

2. Sampling and laboratory techniques

In total, 166 samples for palaeomagnetic, age and geochemical studies have been collected from 11 basic dykes in 6 rock quarries (Fig. 3) and from their baked contacts. In Khmelnik samples from one dyke (site 3) were collected and in the Tomashgorod (Psheli) quarry three dykes (sites 4–6) were sampled. The big Tamashgorod dyke was sampled at two sites (8 and 9) and samples were also collected from a small dyke (site 7) located close to the big dyke at site 8. In Viry and Susly two dykes were sampled from each quarry (sites 10, 11, and 12, 13, respectively).

The sampling was performed using a portable rock coring drill and the samples were oriented with sun and magnetic compasses. The age determinations and chemical analyses were made on specimens from the drill cores used in the palaeomagnetic study.

The intensity and direction of magnetization was measured with a SQUID (2G-DC) magnetometer and the specimens were demagnetized in 5 mT steps up to 40 mT and in 10 mT steps up to 100 mT. Thermal demagnetizations were generally made at temperatures up to 600 °C at intervals of 50 °C up to 500 °C and 10–20 °C for higher temperatures using a Schonstedt (TSD1) oven. The magnetic susceptibility was measured during heating and cooling with a Kappabridge (KLY-3 and CS-3, AGICO).

Components of remanent magnetization were defined using a method of least squares fit of vectors in three dimensional space (Torsvik, 1986; Torsvik et al., 1996) and by principal component analysis (Kirschvink, 1980).

For U–Pb dating of baddeleyite finely ground samples (almost to powder) weighing ca. 0.5 kg were processed from the same drill cores as used for palaeomagnetic measurements. Baddeleyite grains were extracted using a Wilfley water-shaking table following the technique of Söderlund and Johansson (2002). A strong pencil magnet was used to remove magnetic minerals. A small portion of 42 brown baddeleyite grains, with a typical grain dimension of 20–30 μm, was recovered from the limited amount of samples that was available for the geochronology. The best quality baddeleyite grains were handpicked under a binocular microscope and combined into fractions of 5–10 grains each. The grains were transferred using a hand-made micropipette into Teflon dissolution bombs and washed successively in 2–3 N nitric acid and water. A trace of a mixed $^{236-233}U$–^{205}Pb isotopic solution was added. The sample was completely dissolved in HF:HNO_3 (10:1) at 210 °C overnight. Chemical separation of U and Pb resulted in a total procedural blank of 1 pg Pb and 0.2 pg U. U and Pb were loaded together with silica gel onto an outgassed single Re filament and analyzed in a Finnigan Triton thermal ionisation multi-collector mass spectrometer at the Museum of Natural History in Stockholm.

Because of the small amount of baddeleyite, all the Pb isotopes were measured using the peak-switching mode on a Secondary Electron Multiplier (SEM). Initial common Pb was corrected for using the model composition of Stacey and Kramers (1975). The mass discrimination correction for the SEM detector is constant at 0.1% per atomic mass unit. Decay constants are those of Jaffey et al. (1971). All age errors quoted in the text and table and the error ellipses in the concordia diagram, are given at the 95% confidence level. Plotting and age calculations are from Ludwig (2003).

For the U–Pb dating zircons were separated from cruched whole rock samples, each weighing up to 30 kg, using conventional water table, heavy liquid and magnetic techniques. Subsequently handpicked zircons were used for both thermal ionization mass-spectrometry (TIMS) and secondary-ion mass-spectrometry (SIMS) at the Museum of natural History, Stockholm, Sweden.

The TIMS analyses were made on multigrain fractions, two of which represented by elongated prismatic grains and a third represented by short-prismatic to isometric crystals. The latter was abraded to remove external disturbed domains (Davis et al., 1982;

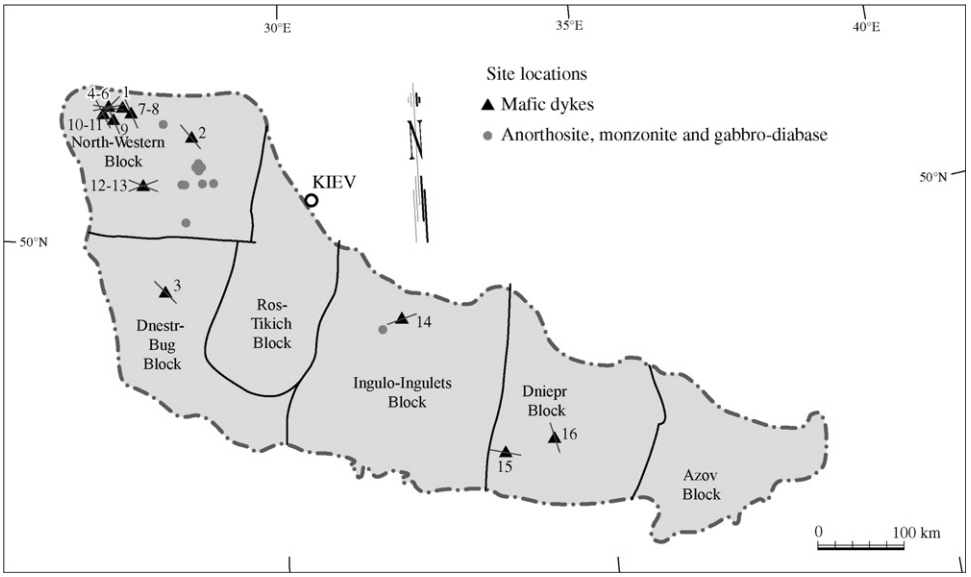


Fig. 3. The crustal blocks of the Ukrainian Shield. The locations of the dykes where samples have been collected for this study and in a previous study (Elming et al., 2001) are marked by filled triangles. Anorthosite, monzonite and gabbro-diabase sites in the study of Elming et al. (2001) are marked by grey dots.

Table 1
Isotope data from TIMS and SIMS analyses.

(A) Method applied	Isotopic ratios				r	Age (Ma) ± 1 σ			Concentration (ppm)	
	$^{206}Pb/^{204}Pb$ $^{207}Pb/^{235}U$	±s, %	$^{206}Pb/^{238}U$	±s, %		$^{207}Pb/^{206}Pb$	$^{207}Pb/^{235}U$	$^{206}Pb/^{238}U$	U	Pb
SIMS 558	6.675	3.6	0.384	3.1	0.86	2043 ± 32	2069 ± 32	2096 ± 55	185.6	92.8
SIMS 5731	6.597	2.0	0.380	1.8	0.93	2040 ± 12	2059 ± 17	2078 ± 33	588.3	264.0
SIMS 3393	5.729	2.1	0.335	1.8	0.87	2013 ± 18	1936 ± 18	1865 ± 30	232.2	102.7
SIMS 1431	4.551	2.4	0.268	1.9	0.78	2005 ± 27	1740 ± 20	1529 ± 25	459.5	156.7
SIMS 595	3.994	2.9	0.246	1.8	0.63	1923 ± 40	1633 ± 24	1417 ± 23	179.8	63.7
SIMS 5305	6.653	1.9	0.378	1.8	0.95	2066 ± 10	2066 ± 17	2067 ± 32	842.0	385.4
TIMS[a] 777	5.644	0.6	0.321	0.6	0.91	2066 ± 5	1925 ± 10	1797 ± 9	322.0	118.8
TIMS[a] 1034	4.955	0.6	0.283	0.6	0.92	2055 ± 4	1812 ± 13	1608 ± 10	675.8	211.82
TIMS[b] 136	6.698	1.0	0.377	0.7	0.73	2084 ± 12	2072 ± 25	2060 ± 17	160.4	95.1
(B) Spot #	Isotopic ratios				r	Age (Ma) ± 1 σ			Concentration (ppm)	
	$^{207}Pb/^{235}U$	±s, %	$^{206}Pb/^{238}U$	±s, %		$^{207}Pb/^{206}Pb$	$^{207}Pb/^{235}U$	$^{206}Pb/^{238}U$	U	Pb
Susly old dyke										
1	6.828	1.1	0.381	1.1	0.95	2097 ± 6	2089 ± 10	2082 ± 19	179.5	84.7
2	4.604	5.8	0.348	1.0	0.18	1546 ± 103	1750 ± 49	1926 ± 17	33.7	22.2
Susly young dyke										
1	2.038	1.1	0.164	1.0	0.90	1430 ± 10	1128 ± 8	978 ± 9	278.7	64.3
2	2.061	2.6	0.135	2.2	0.85	1816 ± 25	1136 ± 18	814 ± 17	642.2	112.3
3	4.517	2.6	0.258	2.5	0.98	2055 ± 10	1734 ± 22	1481 ± 33	280.1	96.0
4	6.686	1.1	0.372	1.0	0.93	2102 ± 7	2071 ± 10	2039 ± 18	140.2	69.5
5	6.697	1.1	0.376	1.0	0.87	2089 ± 10	2072 ± 10	2056 ± 18	99.3	49.1
6	6.769	1.1	0.375	1.0	0.97	2108 ± 4	2082 ± 9	2055 ± 18	684.8	292.0
Aplite vein										
1	6.674	1.0	0.379	1.0	0.99	2067 ± 2	2069 ± 9	2071 ± 18	3172.1	1356.7
2	6.970	1.2	0.386	1.0	0.87	2110 ± 10	2108 ± 11	2105 ± 19	165.4	81.6
3	6.458	1.1	0.371	1.0	0.92	2046 ± 8	2040 ± 10	2034 ± 18	156.2	68.3

(A) U–Pb isotopic data for zircons separated from the Susly granite (sample 05-C1). (B) U–Pb isotopic data for zircons separated from the Susly mafic dykes and aplite vein, SIMS data.

[a] Elongated grains.
[b] Short-prismatic grains, abraded.

Krogh, 1982). The samples were washed using HNO_3, H_2O and acetone, weighed on a microbalance and spiked with a mixed $^{205}Pb/^{235}U$ tracer. Dissolution of zircon was carried out with HF (+HNO_3) in Teflon bombs (Krogh, 1973) at 184 °C. The purified samples were loaded on zone-refined Re filaments with Si-gel and H_3PO_4 and measured on a MAT 262 mass spectrometer on Faraday cups in static mode.

For the SIMS analyses separated zircons and the 91500 zircon standard (Wiedenbeck et al., 1995) were mounted in epoxy resin discs and polished to expose the interior of the zircons. In order to observe the textures of the polished zircons secondary electron microscope (SEM) and CL imaging were performed on a Hitachi SEM. U-Th-Pb geochronological data wrer obtained using the Cameca 1270 ion microprobe at the NORDSIM facility. The Pb/U ratio, element concentration and Th/U ratio calibration was performed relative to the Geostandard zircon 91500. The analytical method follows that described by Whitehouse and Kamber (2005). The common lead correction, when needed, was done using the measured ^{204}Pb signal and modern (i.e. 0 Ma) Pb isotope composition (Stacey and Kramers, 1975). Data reduction employed Excel macros developed by M.J. Whitehouse. Data plotting and age calculations for both SIMS and TIMS results were performed using the program Isoplot (Ludwig, 1999). The results from the isotopic investigation, applying both TIMS and SIMS methods on zircons from the Susly rocks are shown in Table 1.

Palaeomagnetic core samples were also submitted for $^{40}Ar/^{39}Ar$ analysis at the Geochronology laboratory at the University of Alaska, Fairbanks. Grains of hornblende were extracted from one basic dyke. The monitor mineral MMhb-1 (Samson and Alexander, 1987) with an age of 513.9 Ma (Lanphere and Dalrymple, 2000) was used to monitor neutron flux (and calculate the irradiation parameter, J). The samples were heated in a step-wise manner, using a 6 W argon-ion laser and analyzed in a VG-3600 mass spectrometer at the University of Alaska, Fairbanks following the technique described in York et al. (1981), Layer et al. (1987) and Layer (2000). The argon isotopic measurements were corrected for system blank and mass discrimination, as well as calcium, potassium and chlorine interference reactions following the procedures outlined in McDougall and Harrison (1999). Plateau ages were based on the criteria of three or more contiguous fractions (with a roughly constant Ca/K ratio) accounting for more than 50% gas release, having a Mean Square Weighted Deviation (MSDV) less than 2.5. The $^{40}Ar/^{39}Ar$ results are given with all ages quoted to the ±1 sigma level and calculated using the constants of Steiger and Jaeger (1977). Integrated and plateau ages include the error in the irradiation parameter (0.5%).

3. Geochemistry

Radiometric dating of the sometimes relatively thin and fine grained mafic dykes is often difficult. Analyses of their chemical composition may in this respect be of great help, since it may be used to demonstrate the affinity of the dyke and match that to other dykes, which have been well dated. This approach is here used to define groups of dykes with similar geochemical compositions and link those to radiometric ages and palaeomagnetic data.

Dykes in the western part of the Ukrainian shield can be broadly divided into three main groups: (1) those related to the Osnitsk orogenic event with ages of ca. 2000 Ma, which in turn can be tentatively sub-divided into, pre-granite and post-granite subgroups; (2) a suite of Ni-enriched dykes and small intrusions emplaced at ca. 1780 Ma.; (3) a suite of mafic dykes related to the emplace-

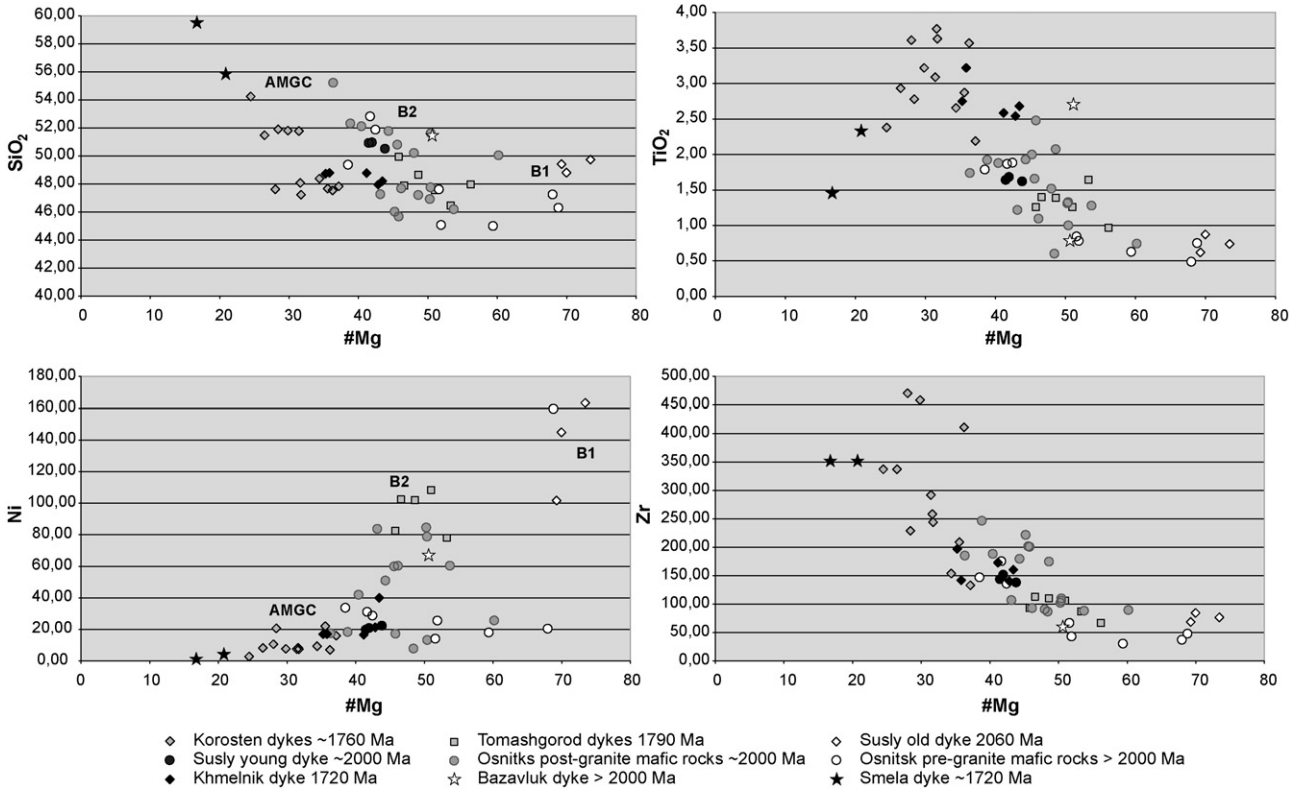


Fig. 4. Geochemical data from the basic dykes of this study together with unpublished geochemical data from other dykes in the north-western Volyn Block. The Viry dyke (10) is plotted as Osnitsk pre-granite dykes (B1), the Tomashgorod quarry dykes (4–6) (sites 4 and 6) as Osnitsk post-granite dykes (B2) and the Bekhi dyke (site 2) as a Korosten dyke (AMGC). Note the gradual increase in Mg# from the anorthosite related dykes to the Osnitsk pre-granite dykes.

ment of the Korosten anorthosite-mangerite-charnockite granite (AMCG) complex. Some of the latter dykes were emplaced as early as 1800 Ma, while the majority was formed at ca. 1760 Ma. These dykes are mainly of basaltic composition, however, some anorthosite related dykes and Osnitsk post-granite dykes plot on the trachy-basalt to basaltic trachy-andesite part of the Le Maitre et al. (1989) diagram.

There are clear geochemical differences between the two groups of Osnitsk mafic rocks. The B1 group (pre-granite mafic rocks, Fig. 4, Appendix A) has high Mg-number (Mg# = Mg/(Mg + Fe)), twice as much MgO compared to mafic rocks of the B2 group (post-granite mafic rocks) and increased concentration of CaO, CO_2, Cr_2O_3, Ni, Co, Cs, Cu (Fig. 4). On the other hand the mafic rocks of the B2 group are enriched in TiO_2, P_2O_5, S, Ba, Ga, Hf, Nb, Rb, Th, U, Zr, Y, REE, Pb, Zn.

According to our geochemical results, the older Susly metadolerite dyke (site 12) belongs to the pre-granite group (B1) of dykes (Table 2), while the crosscutting dyke (site 13) clearly belongs to the post-granite one (B2). Furthermore, the older Susly dyke belongs to a rather rare chemical suite of high-Mg gabbro and only a few analyzed rocks belong to this chemical suite. Most of the mafic rocks related to the Osnitsk orogenic event clearly belong to the

Table 2
Classification of the mafic dykes based on their geochemical composition (Fig. 4, Appendix A) and direction of remanent magnetization (Fig. 9; Table 4). EX denotes intrusions simultaneous with early evolutionary stages of the AMCG Korosten complex. The remanence directions of the sites 8 and 9 fall within the palaeomagnetic group II, however, since the remanence probably is of viscous origin a question mark is put for palaeomagnetic grouping of these sites.

Geochemic (X) and palaeomagnetic (I–III) classification of the mafic dykes				
Locality (site)	AMCG	Post-granite (B2)	Pre-granite (B1)	Paleom. group
Bekhi (2)	X			I
Khmelnik (3)	X			III (Rev. I)
Tamashgorod Quarry (4)		X		II
Tamashgorod Quarry (6)		X		II
Tamashgorod dyke				
Small dyke (7)		X		II
Big dyke (8)		EX		II?
Oleksandivka (9)		EX		II?
Viry (10)			X	II
Susly older dyke (12)			X	III
Susly younger dyke (13)		X		II
Smila (14)	X			I
Bazavluk (15)			X	III

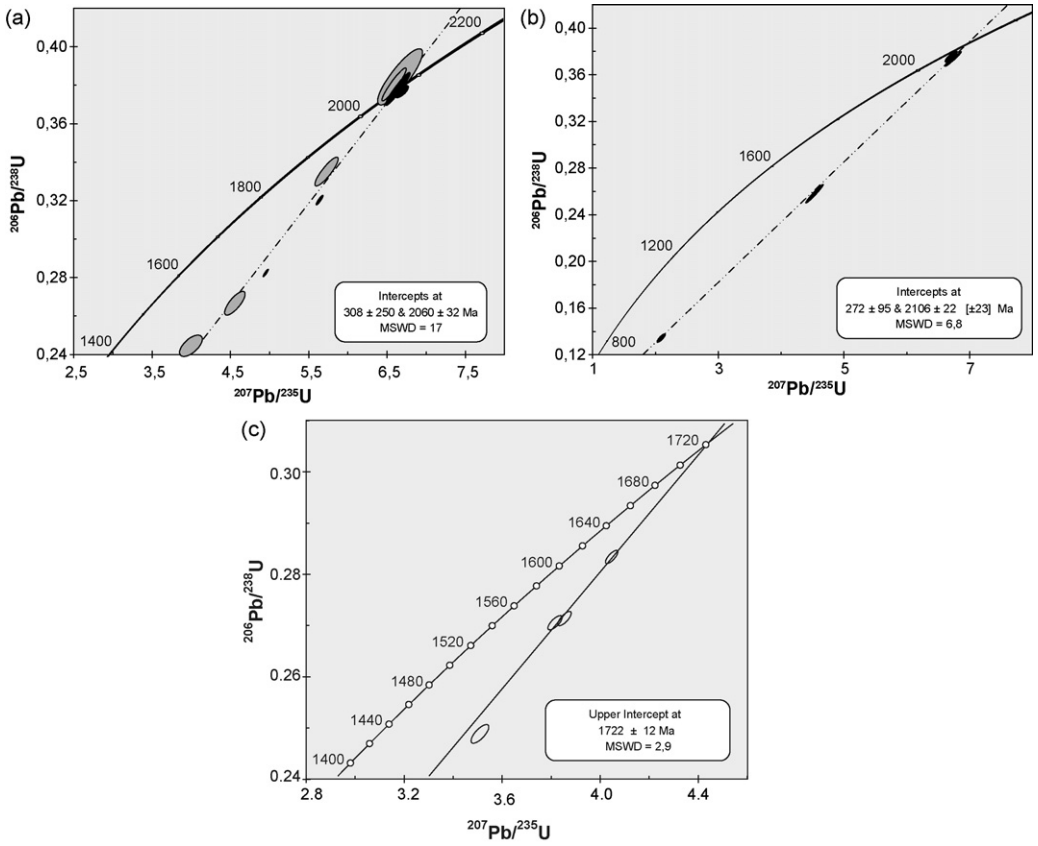


Fig. 5. (a) U–Pb concordia diagram for zircon separated from the granitic host rock at Susly. (b) The U–Pb concordia diagram for zircon separated from the younger Susly Dyke (site 13). The zircons are most probably inherited. (c) U–Pb Concordia diagram for baddeleyite separated from the Khmelnik dyke (site 3). A forced lower intercept of 600 ± 100 Ma is used based on the assumption of Pb loss that occurred during thermal heating and late Proterozoic volcanic activity.

pre-granite group and appear as rather small root-less metamorphosed and granitized bodies hosted by granite. However, there are important exceptions from this subdivision based on geochemical data, here represented by the dyke (site 10) of amphibolized dolerite that outcrop in the Viry open pit. This dyke possesses high Mg# but appear to cut the Osnitk granite, i.e. geochemically this dyke belongs to the B1, pre-granite group, but geologically it is post-granitic.

The dykes at sites 4, 6 and 7 are attributed to the Osnitsk post-granite group, while the Bazavluk dyke (site 15) in the Dniepr domain has a composition similar to the dykes of the pre-granite group. However, since this dyke is located in a different geological environment it may not necessarily be related to the pre-granite Osnitsk orogenic event.

The AMCG (or anorthosite-rapakivi granite) related dykes (Bekhi (site 2) and Smila (site 14), Fig. 3) belong to the suite of moderately fresh (sometimes heavily altered) ilmenite and apatite bearing dykes. These are characterized by low Mg#: generally 37–24, but in the Smila dyke as low as 21–17 (Fig. 4, Appendix A), high TiO_2 content (usually above 2.50%), high K_2O content (usually above 2.00%, which reflects the jotunitic nature of these dykes), high P_2O_5 content (usually above 0.50%). In contrast, the concentration of all other main oxides is relatively low. AMCG related dykes are rich in incompatible elements (apart from Sr) and depleted in Cr, Ni, Cu, V. The composition of the Khmelnik dyke (site 3) is very close to that of the Bekhi and Smila dykes and probably represents the same tectonic event.

Ni-enriched dykes were formed simultaneously with early evolutionary stages of the Korosten complex. In contrast to the latter, initial melts of Ni-enriched dykes and small intrusions were formed due to melting of the depleted mantle source. These dykes, represented here by the Tomashgorod dyke (sites 8 and 9), are fresh pigeonite or olivine dolerites, while the small layered intrusions are composed of peridotite, gabbro and anorthosite. Ni-enriched dolerites possess moderately high Mg# number (56–47), rather low TiO_2 content (below 1.50%, Fig. 4), Na_2O (below 3.00%), K_2O (below 1.00%), and P_2O_5 (usually below 0.20%). At the same time these rocks are enriched in Al_2O_3 (above 15.00%) and CaO (above 8.00%). With respect to trace elements, these rocks are enriched in Ni (typically above 80 ppm), V and Cu, while concentration of incompatible elements is typically low.

4. Radiometric age determinations

The host granite of the Susly quarry carries a large amount of zircons that are represented by two morphological types: (1) short-prismatic and (2) elongated. The grain size is about 0.1 mm, and both varieties of zircon are transparent and slightly rose-coloured. The elongated crystals are better developed and often have sharp bi-pyramid tips, while the short-prismatic crystals display smoothed rounded tips with poorly developed bi-pyramids. Both varieties are almost free of inclusions. In cathode luminescence (CL) images both reveal “lengthwise” zoning and transverse sections appear oscillatory.

There are some evident discrepancies between results obtained by TIMS and SIMS methods for the Susly granite (Table 1). Specifically, three TIMS analyses of multigrain fractions form a linear array that intersects concordia at 2081.8 ± 8.7 Ga (lower intercept at 163 ± 57 Ga, MSWD = 0.57). However, a regression including all six SIMS analyses yielded an age of 2045 ± 29 Ma (lower intercept at 354 ± 220 Ma, MSWD = 5,1). Plotting all nine results including both TIMS and SIMS on one concordia diagram, yields an age of

Table 3
U–Pb TIMS data of baddeleyite from the Khmelnik dolerite dyke.

Sample	Fraction[a] (n)	U/Th	Pbc/Pbtot[b]	$^{206}Pb/^{204}Pb$ raw[c]	$^{207}Pb/^{235}U$ [corr][d]	±2s, % err	$^{206}Pb/^{238}U$	±2s, % err	$^{207}Pb/^{235}U$ [age, Ma]	$^{206}Pb/^{238}U$	$^{207}Pb/^{206}Pb$	±2s, % err	Concordance
SK	Bd-a (5)	7.8	0.147	373.6	3.8562	0.58	0.27127	0.43	1604.6	1547.3	1680.6	7.1	0.88
	Bd-b (8)	2.3	0.225	216.7	3.5057	0.86	0.24836	0.61	1528.5	1430.0	1667.6	10.7	0.80
	Bd-c (10)	4.0	0.193	264.6	3.8167	0.63	0.27045	0.43	1596.3	1543.1	1667.2	7.9	0.87
	Bd-d (6)	9.4	0.159	339.5	4.0510	0.52	0.28351	0.38	1644.5	1609.0	1690.2	6.2	0.91

[a] Bd: baddeleyite, *n*: number of grains in each analysis.
[b] Pbc: common Pb; Pbtot: total Pb (radiogenic + blank + initial).
[c] Measured ratio, corrected for fractionation and spike.
[d] Isotopic ratios corrected for fractionation (0.1% per amu for Pb), spike contribution, blank (1 pg Pb and 0.2 pg U), and initial common Pb. Initial common Pb corrected with isotopic compositions from the model of Stacey and Kramers (1975) at 1720 Ma.

2060 ± 32 Ma (lower intercept at 308 ± 350 Ga, MSWD = 17, Fig. 5a). This age is similar to a single concordant analysis of 2067 Ma, which is probably the most representative of the Susly granite crystallization.

The older Susly metadolerite dyke in the Susly quarry carries a very limited amount of zircon grains. A CL investigation revealed the presence of a corroded (inherited) core in one zircon. This core yielded a concordant age of 2095 ± 5 Ma, while the age of the wide rim is discordant and has a $^{208}Pb/^{232}Th$ age of 2080 Ma. However, both ages are higher than the age of the host rock and must be regarded with caution.

The younger Susly metadolerite dyke carries a large number of zircons that can be subdivided into three groups. The first group includes prismatic slightly resorbed brown crystals with well developed pyramidal tips. The cathode luminescence (CL) is rather weak and reveals some fuzzy zonality. The second group includes grey prismatic crystals with heavily resorbed tips. One of the grains differs by its elongated shape. CL images of crystals from this second group reveal weak luminescence with concentric zonality and inherited cores are absent. Finally, the third group of grains consists of rather fine (<100 μm) prismatic and isometric euhedral colourless transparent crystals with well developed pyramidal tips. In CL images crystals from this group display concentric zonality. Six zircon grains that represent the three groups of zircons have been analysed. Three of the zircons are almost concordant and their weighted average age is 2105 ± 7 Ma. One of the zircons is very discordant and falls far away from the other zircons in the U–Pb isotopic plot and it deviates significantly from the regression line.

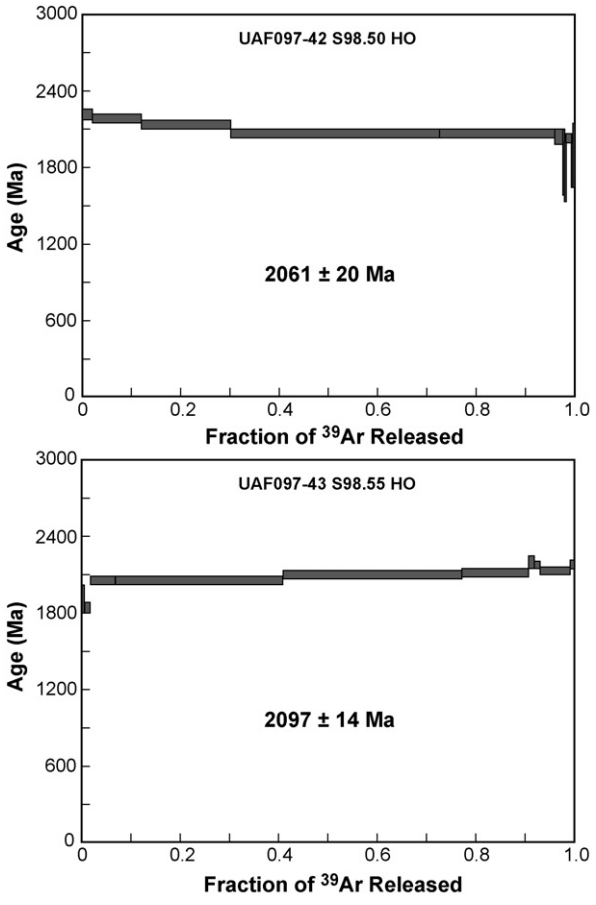


Fig. 6. $^{40}Ar/^{39}Ar$ dating of hornblende separated from the old dyke in Susly. Samples were taken from the central part (U98.55) and at ca. 0.5 m (U98.50) from the contact to the host granite of this ca. 25 m wide dyke and both datings meet the criteria for a plateau age.

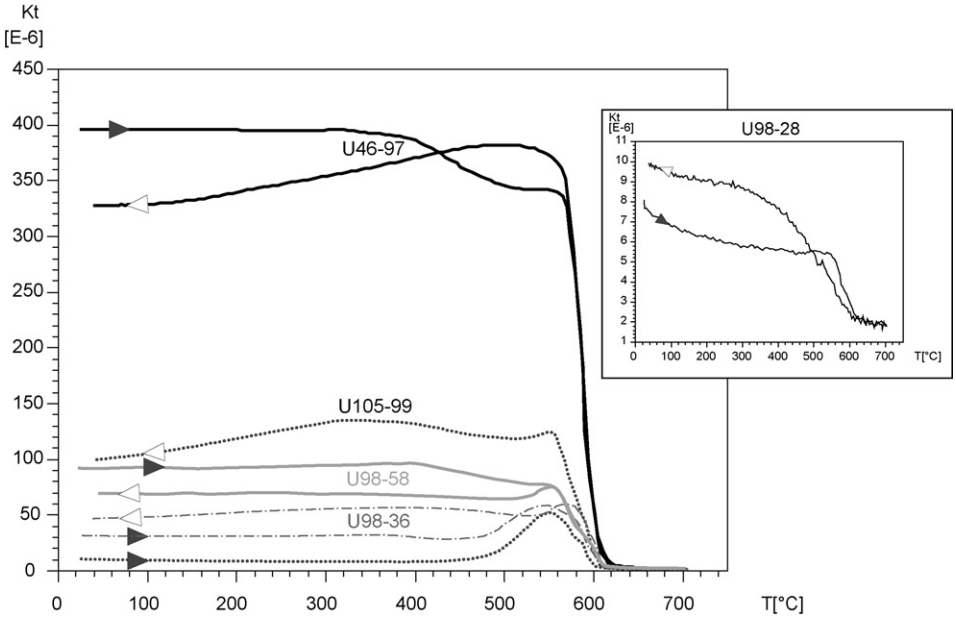


Fig. 7. Examples of susceptibility-temperature measurements for the Khmielnik (site 3, U46-97), young Susly (site 13, U105.99), old Susly (site 12, U98-58), Tomashgorod quarry site 5 (U98-36) and site 6 (U98-28) dykes. The magnetization is generally carried by magnetite, as noted by the Hopkinson peak just below the Curie temperature of magnetite, and to some extent by hematite (e.g. U46-97 and U98-28).

A regression line based on the five other grains intercepts Concordia at 2106 ± 22 Ma (Fig. 5b). There is no visible correspondence between zircon appearance and position on the U–Pb isotopic plot.

The ages of the zircons separated from both dykes correspond to ages of zircons commonly found in the Tetriv series of metasediments (Scherbak et al., 2008), which envelope the Susly granite massif and they are therefore most probably inherited.

A thin aplite vein that cuts the younger Susly dyke contains an abundance of zircons. They usually appear as fine-grained (<50 μm) light-pink transparent well-shaped subhedral crystals. One of the grains was abnormal due to its elongated (up to 500 μm) bent shape. CL images demonstrate sharp concentric zonality with wide dark rims. Three grains were dated; all of them possessed concordant ages of 2046 ± 8, 2067 ± 2, and 2110 ± 10 Ma, respectively. These zircons are probably inherited from the Susly granite and the Tetriv metasediments.

The samples collected for $^{40}Ar/^{39}Ar$ dating of the older dyke in Susly were taken from the central part (U98.55) and at ca. 0.5 m (U98.50) from the contact to the host granite of this ca. 25 m wide dyke. Both analyses of the hornblende samples meet the criteria for plateaux and well defined ages of 2097 ± 14 Ma (U98.55) and 2061 ± 20 Ma (U98.50; Fig. 6) were obtained.

The four analyses of baddeleyite grains separated from the Khmelnik dyke (site 3) plot significantly below the concordia line and define an imprecise upper intercept age of 1694 ± 51 Ma with a relatively high MSWD value of 5.2. If the most discordant fraction (also associated with the highest common Pb/total Pb ratio, Table 3) is excluded and if it is assumed that Pb loss occurred during thermal heating related to the formation of the Volyn, continental flood basalt province, i.e. forcing a lower intercept through 600 ± 100 Ma (e.g. Elming et al., 2007; Compston et al., 1995), it yields an upper intercept becomes 1722 ± 12 Ma (MSWD = 2.9, Fig. 5c). This result is very close to the age of the anorthosite complexes (site 13 and related dyke site 14 in Elming et al., 2001) in the Ukrainian Shield (Amelin et al., 1994; Scherbak et al., 1995; Shumlyanskyy et al., 2008a,b) and it is interpreted as the age of the Khmelnik dyke although an uncertainty in this interpretation is admitted.

5. Palaeomagnetic and rock magnetic results

The remanence directions isolated during AF and Thermal demagnetizations usually agree and are often combined when calculating the site means, however, a slight majority of the data (ca. 55%) are from thermal treatment.

In the Novograd-Volynskyy area (Fig. 2) there are differently trending dykes and in one quarry at Susly two dykes cross cut each other. The magnetic mineral of the older dyke (site 12) in the Susly quarry, as indicated from susceptibility-temperature measurements, is magnetite-titanomagnetite (Fig. 7) and thermal demagnetization also reveals magnetite as the mineral carrying the remanent magnetization (Fig. 8f). The coercivity of remanence is generally high with a median destructive field (MDF) of 10–70 mT. The mean remanence direction of this dyke (site12; decl. = 211.5°, incl. = 40.4°, α_{95} = 4.8°; Table 4) is close to the mean remanence direction of the baked granite (sampled up to 0.5 m away from the dyke; decl. = 203.0°, incl. = 28.3°, α_{95} = 9.7°, Fig. 9b), which is isolated both in alternating field and thermal demagnetizations. No stable remanence could be isolated in the granitic host rock sampled far from the dyke. The dyke (site 13) in Susly, which cuts the dyke at site 12 is sampled some 70 m from the crossing. It has a remanent magnetization of low coercivity (MDF often <5 mT), which means a large portion of the remanence is likely of viscous origin. Only in a few specimens (e.g. Fig. 8g) can stable remanences be isolated in alternating field treatment. The dominating magnetic mineral is magnetite as indicated by the susceptibility–temperature measurements, where a Hopkinson peak appears just below the Curie temperature for magnetite (Fig. 7). The Tub spectra of the remanent magnetization are generally wide (Fig. 8g) but consistent directions are identified close to the Curie point of magnetite. The mean remanence direction (decl. = 47.3°, incl. = 64.8°, α_{95} = 4.2°) is very different from that of the old dyke and significantly different from that of the International Geomagnetic Reference Field (IGRF: decl. = 5.8°, Incl. = 67.5°).

Three dykes crop out in the Tomashgoorod quarry. The remanent magnetization of the ca. 0.5 m thick basic dyke (site 4) is carried by magnetite. The coercivity of remanence is high (MDF ca. 15–20 mT), and the remanence is unblocked over a narrow temper-

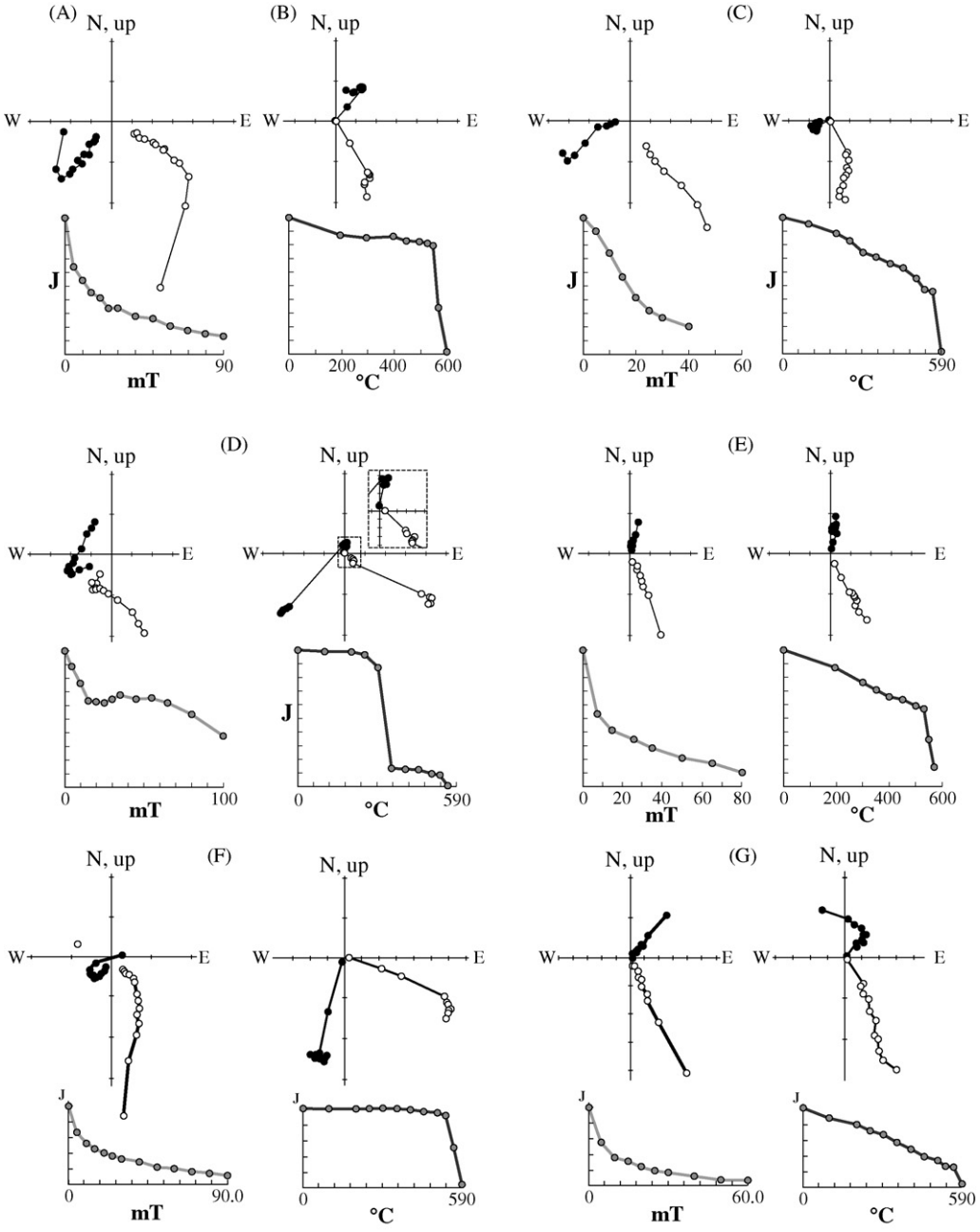


Fig. 8. Exemples of remanence behaviour during alternating field and thermal demagnetizations for the Khmelnik site 3 (A), Tomashgorod quarry dyke site 4 (B); Tomashgorod quarry dyke site 5 (C); Tomashgorod quarry dyke site 6 (D); Viry dyke site10 (E); older site 12 (F) and younger Susly dyke site 13 (G). Note the big drop in intensity and change in remanence direction at ca 300 °C during thermal demagnetization of the Tomashgorod quarry dyke site 6.

ature range at temperatures >550 °C (Fig. 8b). The mean remanence direction (decl. = 31.5°, incl. = 50.1°, α_{95} = 5.0; Table 4; Fig. 9a) is similar to that of the younger dyke (site 13) in Susly. From samples of the 4.5 m thick basic dyke (site 5), the susceptibility–temperature measurements (Fig. 7) suggest that the magnetization is carried by magnetite. A characteristic remanence is isolated at temperatures close to the Curie temperature of magnetite, however the Tub range is rather wide (Fig. 8c). The site mean direction (decl. = 229.4°, incl. = 51.4°, α_{95} = 4.8) is close to that of the older dyke in Susly. The ca. 0.5 m thick dyke (site 6) in this quarry carries two components of remanent magnetization and they are best defined by thermal demagnetization (Fig. 8d). A low Tub component (decl. = 230.0°, incl. = 27.9°, α_{95} = 9.1°, Fig. 9a) carrying 40–90% of the NRM is found unblocking at a temperature between 300 and 350 °C. This component is of high coercivity and isolated in fields higher than 40 mT. A high Tub component (decl. = 15.0°, incl. = 37.8°, α_{95} = 4.6°) is defined in a rather narrow temperature range around 560 °C. This component is identified in fields between 20 and 60 mT, i.e. in fields lower than those of the low Tub component. Susceptibility–temperature measurements reveal magnetite and possibly also hematite to be carrier of magnetization.

In the Viry quarry a 2.9 m thick dyke (site 10) carries a characteristic magnetization (decl. = 18.7°, incl. = 59.7°, α_{95} = 4.7°), similar to that of site 4 in the Tomashgoorod quarry, and isolated in alternating fields >25 mT and Tub >520 °C (Fig. 8e). The unblocking temperature range is rather wide and 40–60% of the NRM is erased at temperatures below 520 °C. The baked granite contact to the dyke (site 10b) carries a remanence direction close to that of the

Table 4
Palaeomagnetic data for the basic dykes of this study and of Elming et al., 2001. Glat/Glon = geographic latitude and longitude of site locality, (site) = site number and remanence component. (HHc and HTb (LTb) stands for high (low) coercivity and unblocking temperature, respectively), *n* = number of samples, Decl. (incl.) = mean declination (inclination) of the site, α_{95} = semi- angle of the cone of 95% confidence, *K* = Fisher precision parameter, Plat. (Plon.) = latitude (longitude) of the (Virtual) palaeomagnetic poles, Dp (Dm) = semi-axes of the 95% confidence ellipsoid of the pole position, A_{95} = semi-angle of the 95% confidence circle of the pole position, Age = radiometric age. TG = Tamashgorod dyke. The rotated poles are rotated with respect to rotation parameters: Lat. = 48°N, Lon. = 42.2°E, Rot. Angle = 43°.

Ukrainian basic dykes

Latitude/Longitud	Locality (nb)	*n*	Decl.	Incl.	α95	*k*	Plat.(°N)	Plon.(°E)	Dp/Dm	A95	Age (Ga)	Comments
NW Domain												
51.30/27.23	Rakitny (1)	10	45.5	53.7	7.8	24	53.3	126.9	7.6/10.9	9.1	Ca 2.0	Elming et al., 2001
50.97/28.48	Bekhi (2)	17	33.2	1.1	3.8	88	32.3	168.1	1.9/3.8	5.1	Ca 1.77	Elming et al., 2001
	Baked cont. (2b)	7	39.6	0.2	10	38	29.1	161.6	5.0/10.0	7.1		
	Host rock	5	21.6	35	25	10	54	172.2	16.6/28.8	21.9		
49.52/28.02	Smielnek (3)	24	217.1	36.1	4.8	39	13.2	172.5	3.2/5.6	4.2	1.72	This work
	Tomash quarry											
51.32/27.05	A:HHc, HTb (4)	14	32.3	48.7	4	90	57.7	246.7	3.5/5.3	4.3		This work
	B: HTb, HHc(5)	17	229.4	51.4	4.8	55	−4	166.9	4.4/6.5	5.3		
	C:HTb (6a)	11	15	37.8	4.6	98	57.7	180.2	3.2/5.4	4.2		
	C:LTb (6b)	9	230	27.9	9.1	33	10.9	158.1	5.4/10.0	7.3		
	Tomashgorod dyke											
51.20/27.40	TG-Small dyke (7a)	5	21.8	58.5	14.6	28	70.6	146.9	16.1/21.7	18.7		This work
	TG-Small HHc (7b)	5	226.4	26.7	17.3	29						
	TG-Big dyke (8)	7	354	47	8.7	49	66.6	220.6	7.3/11.2	9		
51.17/27.15	TG-Alexandrovka (9)	9	5	47.5	11.2	22	67.1	195.9	9.5/14.6	11.8		This work
51.21/26.91	Viry, A (10)	19	18.7	59.7	4.7	53	73.3	149.3	5.3/7.1	6.1		This work
	Baked cont.(10b)	3	17.3	72.2	12	107	78.3	79.1	18.7/21.2	19.9		
	Small dyke, B (11)	3	18.9	43.2	17.3	52	60.2	170.8	13.3/21.5	16.9		
50.55/27.63	Susly old (12)	21	211.5	40.4	4.8	45	11.3	178.7	4.2/6.8	5.3	2.07	This work
	Baked cont.(12b)	4	203	28.3	9.7	91	21.3	183.7	5.8/10.6			
	Young dyke (13)	20	47.3	64.8	4.2	61	59.1	106	5.4/6.8	6.1		
Ingulo-Ingulets Domain												
49.19/31.86	Smila (14)	18	45.6	−11.8	4.5	59	22.1	161.8	2.3/4.6	5.1	ca 1.72	Elming et al., 2001
Dniepr Domain												
47.80/34.10	Bazavluk (15)	16	205.9	44.1	3.8	95	12.8	190.3	3.0/4.8	5.7	1.8–2.1?	Elming et al., 2001
47.96/33.24	Kirovorok (16)	15	43.1	−0.4	3.9	96	29.1	161.8	2.0/3.9	4.4		Elming et al., 2001

Mean Pole Positions

			Mean Pole			Rotated Pole		
			Plat.(°N)	Plon.(°E)	A95	Plat.(°N)	Plon.(°E)	Age (Ga)
A	12, 12b, 15	3	15.7	182.9	13.7	24.8	225.9	ca. 2.07
B	1,4,6a,10,10b,13 site 1 in Elming et al.	7	64.4	140.4	12.8	78.6	263.6	ca. 2.0
C	2,2b,16 sites 4,5,9,10,11 in Elming et al.	8	28.8	169.8	4.4	42	221	1.76–1.77
D	sites 6,7,8 in Elming et al.	3	20.5	167.4	5.6	35.8	213.3	ca. 1.74
C + D		11	26.5	169.1	3.9	40.3	218.7	1.74–1.77
E	3,5,6b sites 13,14 in Elming et al	5	10.7	163.2	10.2	28.7	204	ca. 1.72
Sandstone	site 3 in Elming et al.	1	39.2	172.5	9	49.3	232.1	1.8–2.0

dyke (Table 3) and so does that of a thin ca. 0.5 m thick neighbouring dyke (site 11). However, no stable remanence could be isolated in the unbaked host rock, which means no full field test could be demonstrated.

Samples from a ca. 200 m thick dyke, the Tomashgorod dyke (site 8), carry a NRM of low coercivity with MDF of <5 mT. A fairly stable but probably viscous remenance component is isolated in low alternating fields and the calculated mean direction (decl. = 354.0°, incl. = 47.0°, α_{95} = 8.7°) is not very different from the direction of the IGRF. In one sample a stable remanence of high coercivity, but with a significantly different direction (decl. = 243.7°, incl. = 45.1°), survives alternating fields up to 70 mT. A ca 3.0 m thick dyke (site 7), sampled some 130 m from the big dyke, carries a high coercivity magnetization with a mean direction (7b; decl. = 226.4°, incl. = 26.7°, α_{95} = 17.3°) similar to the high coercivity component traced in one sample from the big dyke.

The big Tomashgorod dyke has also been sampled at another site (9) and here again the coercivity of the NRM is low and the mean remanence direction (decl. = 5.0°, incl. = 47.5°, α_{95} = 11.2°) is not very different from the direction of the IGRF. The Tub range is wide and 50% of the remanence intensity is erased at a temperature below 400 °C. In a few samples a fraction of the NRM, with a significantly different direction (decl. = 227.6°, incl. = 18.8°), is stable up to 650 °C. This direction resembles those of high coercivity remanence isolated at the other site of the Tomashgorod dyke.

The basic dyke in Khmelnik (site 3) is ca. 25 m thick. A stable remanent magnetization is defined after demagnetization in alternating fields higher than 50 mT (Fig. 8a). The Tub range is generally rather narrow with ca 60 to 80% of the intensity unblocking at temperatures between 530 and 560 °C. Susceptibility–temperature measurements indicate that the dominating magnetic mineral is

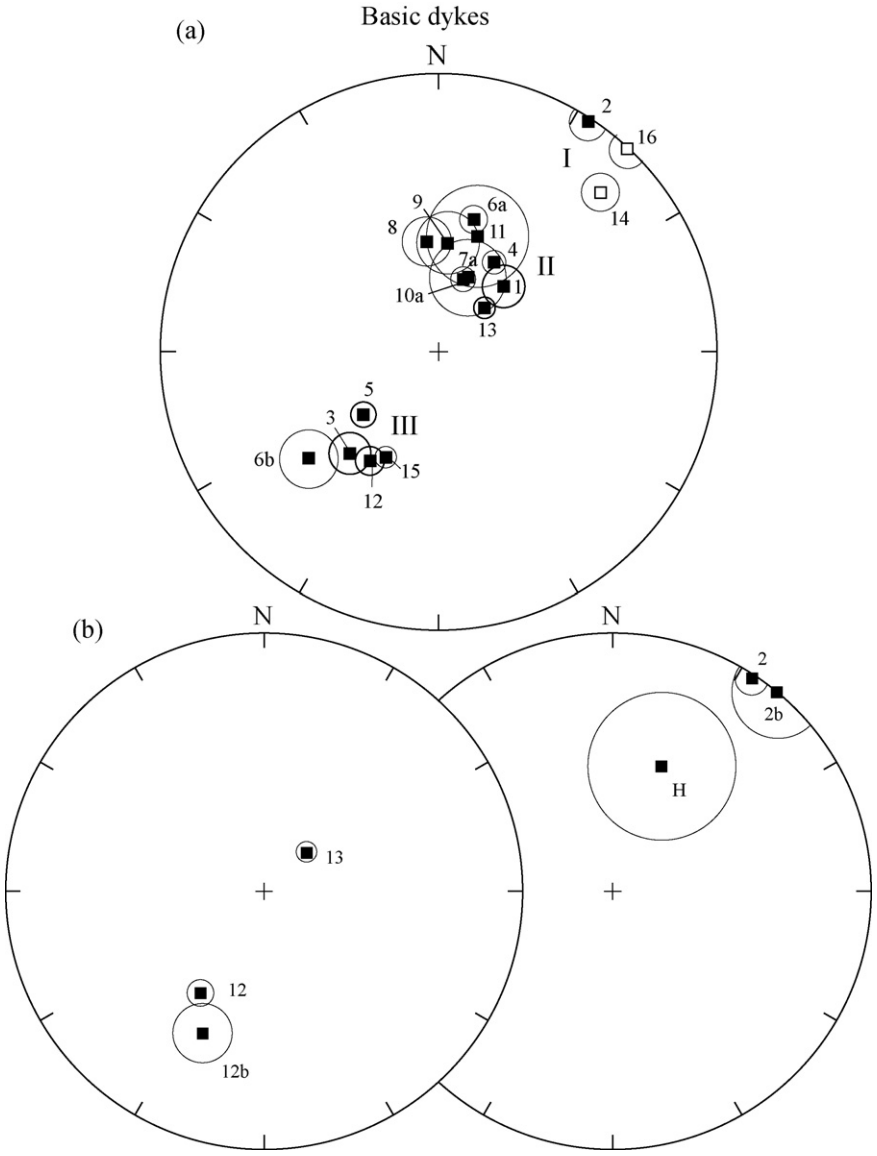


Fig. 9. (a) Site mean remanence directions for the basic dykes of this study (Table 4) and from Elming et al. (2001) plotted with their 95% confidence limits. Filled (open) symbols denote positive (negative) inclinations. The site means form three groups of directions (I) shallow northeast, (II) intermediate northeast and (III) intermediate southwest. From the geochemical analyses (Fig. 4) the group I dykes together with the reversed magnetized dykes at sites 3, 5 and 6b, can be referred to as the AMCG set of dykes, the group II apart from the sites 10 and 11 as the B2 set of dykes and sites 12 and 15 of group III as the B1 dykes (Table 2). (b) Baked contact tests demonstrated for the older dyke in Susly (sites 12 and 12b; Table 3) and the Bekhi (sites 2 and 2b) dyke. The remanence direction of the cross cutting younger Susly dyke (site 13) is significantly different from that of the older dyke. The host rock direction at Bekhi is denoted by H.

magnetite (Fig. 7) and high Hc and Tub components form a site mean direction at decl. = 220.6° and incl. = 38.4° (α_{95} = 6.9°; Table 4; Fig. 9a).

A stable high coercivity remanence was isolated in samples from the host rock in the Bekhi quarry. The samples were collected more than 100 m from the 6 m thick dyke (site 2) and here we add the result from one more sample and combine it with the results from Elming et al. (2001) to form a mean at decl. = 21.6, incl. = 35.0 (Table 4). Even if this mean is poorly defined (α_{95} = 25.0) it is significantly different from both the remanence direction of the dyke and its baked contact, which means a positive baked contact test is demonstrated. The samples of the baked contact were collected up to 2 m from the dyke.

The site mean remanence directions cluster in three groups (Fig. 9a), one in shallow positive and negative inclinations and northeast declinations (I), another in intermediate to high inclination and northeast (II) and a third in intermediate positive and south-western declinations (III), most probably representing different generations of dyke intrusions.

6. Discussion

The grouping of dykes with reference to the palaeomagnetic data (Fig. 9a) correlates rather well with the grouping based on the geochemical data (Table 2). This means palaeomagnetic group I can be referred to as AMCG dykes and group II can generally be referred to as Osnitsk post-granite dykes (B2), however, the Viry dyke (site 10) has a composition corresponding to the Osnitsk pre-granite dykes. Palaeomagnetic group III is a mixture of two generations of dykes, one represented by the 2060 Ma Susly dyke of Osnitsk pre-granite affinity and the other by the ca. 1720–1760 Ma AMCG related dykes with reversed polarity of remanence. The Khmelnik dyke is an example of such a reversed magnetized dyke. This dyke has here been dated at 1722 Ma, which is somewhat younger

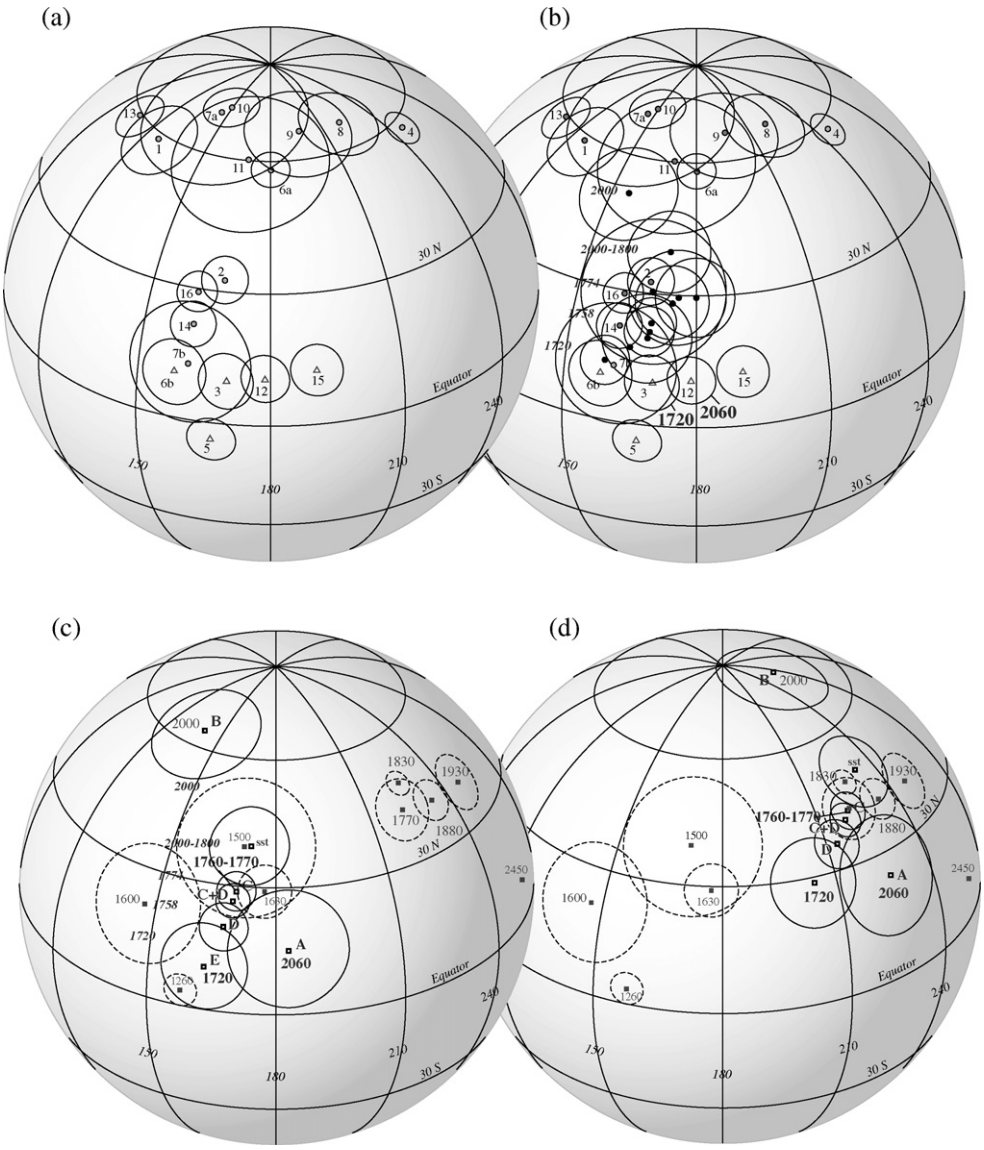


Fig. 10. (a) The palaeomagnetic pole positions calculated from the site mean directions of the dykes. Open triangles denote poles calculated from the group lll dykes (see Fig. 9. (b) The palaeomagnetic poles of the dykes plotted together with pole determined from the magnetization of anorthosites, monzonite and sandstone in the north-western Volyn Block and the Ingulo-Ingulets Block (Elming et al., 2001). The ages determined here for the dykes 3 and 12 are shown in large bold numbers. (c) Palaeomagnetic poles and VGPs calculated from the mean of sites means of basic dykes and site means of coeval anorthosites (Table 4). The confidence circles for the Fennoscandian poles (Table 5) are shown with hatched thin lines. Note the miss-fit between the coeval poles of the Ukraine and Fennoscandia (Table 5). (d) Ukranian poles rotated (Rotation parameters: Lat. = 48°N, Lon. = 42.2°E, Rot. angle = 43°) to the Fennoscandian poles. Note the similar trend in apparent polar wander for the two shields after ca. 1.8 Ga, but the significant difference in trend for older ages.

than the AMCG complexes, but geochemically it is very close to the Bekhi and Smila dykes, which probably signify the same tectonic event. The Smila dyke is related to the Korsun-Novomirgorod complex, which was suggested to be emplaced at ca. 1720 Ma (Scherbak et al., 1995), however new data indicate an age of ca. 1750 Ma (Shumlyanskyy et al., 2008b), i.e. somewhat younger than the Korosten complex. The calculated pole position for the Khmelnik dyke (3) is similar to that of the Smila dyke and the geochemical signatures and ages suggest that the Khmelnik dyke (site 3), the Smila (site 14) and the Bekhi (site 3) dykes all belong to the AMCG group of dykes.

The older dyke in the Susly quarry carries a characteristic magnetization for which a positive baked contact test is demonstrated (Fig. 9b). The ca. 2060 Ma magnetization age of the dyke is estimated from the $^{40}Ar/Ar^{39}$ hornblende ages, one of which (2061 ± 20 Ma) is very close to the proposed 2067 Ma crystalline age of the Susly granite. The 2095 and 2106 Ma U–Pb zircon ages of the older and younger Susly dykes, respectively, probably represent the ages of inherited zircons. The palaeomagnetic pole calculated from the mean direction of the ca. 2060 Ma old dyke (Plat. = 11.3°N, Plon. = 178.7°E) is not far from the pole calculated for the dyke at Bazavluk (site15). That dyke has not been dated, but Mikhailova et al. (1985) reported ages in the interval 1800–2100 Ma for dykes in the Dniepr Block, which may argue for the age of the Bazavluk pole (Plat. = 12.6°, Plon. = 190.3°, A_{95} = 5.7°) being in this range. Geochemically the dyke plot within the group of Osnitsk pre-granite dykes, but the composition is different from that of the old Susly dyke. A mean pole position (Plat. = 15.7°, Plon. = 182.9°, A_{95} = 13.7°), representing an age of ca 2060 Ma, is calculated from the site means 12, 12b and 15 (A; Fig. 10c, Table 4). This pole fulfils most of the quality criteria of Van der Voo (Q = 5; Van der Voo, 1990), i.e. (1) the dyke is well dated, (2) the number of samples (n = 41) is sufficient,

Table 5
Reference poles for the Fennoscandian Shield.

Rock	Plat (°N)	Plon (°E)	A95	Age (Ga)	Reference
Mean Baltica					
Post Jotnian Sills	4	158	4	1.26	Pesonen et al. (2001)
Mean Baltica					
Swedish rapakivi intrusions	39	171	16	1.51	Pesonen et al. (2001)
Mean Baltica					
Basic dykes	29	177	6	1.63	Pesonen et al. (2001)
Ropruchey-Shosksha fm	42	221	7	1.77	Pisarevsky and Sokolov (2001)
Haukivesi dykes	48	225	3	1.83	Neuvonen et al. (1981)
Mean Baltica					
gabbro, gabbor-diorite	41	233	5	1.88	Pesonen et al. (2001)
Tsuomasvarri					
gabbro-diorite	40	247	6.8	1.93	Mertanen and Pesonen (1994)
Russian Karelian dykes	10	256		2.45	Mertanen et al. (1999)

(3) adequate demagnetization, (4) a positive field test is demonstrated, (5) tectonic coherence with craton. I also satisfy those of Buchan et al. (2000) for a key pole, i.e. (1) well defined age of magnetization, (2) primary origin of magnetization is demonstrated, (3) adequate demagnetization. However, since secular variation is probably not averaged out the pole may be regarded as a VGP.

The pole determined for the Viry dyke (site 10) is not very different from the pole of Rokitne gabbro-diabase (site 1) for which a magnetization age has been estimated at ca. 2000 Ma (Elming et al., 2001). A 2000 Ma mean palaeomagnetic pole (B; Plat. = 64.4°, Plon. = 140.4°, A_{95} = 12.8°; Table 4; Fig. 10c) is calculated from the site means of sites 1, 4, 6a, 10a, 10b, 13 combined with site 1 in Elming et al. (2001). For this mean pole secular variation is averaged out and a primary origin of the magnetization may be indicated from the distance (ca. 100 km) between the sites with consistent poles, and with no signs of metamorphism (Buchan et al., 2000). However, the age of magnetization is not well defined.

Ediacaran volcanic activity in the western margin of the East European Craton (Bialowoska et al., 2002) can also be a source of remagnetization. The remanence directions isolated in basalts from the north-western Volyn domain are however significantly different from the directions defined in the basic dykes (Elming et al., 2007).

Since the remanence of the 1790 Ma Tomashgorod dyke (1790 ± 4.4 Ma, U–Pb, zircon; Shumlyanskyy et al., 2008a) is likely of viscous origin, the site means of 7a, 8 and 9 are not included in the calculation of mean pole positions. The small dyke (site 7b), with a different chemical composition, is probably older and remagnetized during the intrusion of the big dyke (site 7a) and it is not used in the mean calculations or plotted in Fig. 9.

A positive baked contact test is also demonstrated for the Bekhi dyke (Fig. 9b), which is related to the Korosten anorthosite Complex. Other basic dykes related to the Korosten Complex have been dated at 1761 ± 25 Ga (U–Pb, zircon; Verhogliad, 1995). The pole position of this dyke is not significantly different from the pole of the Kryvyy Rog dyke (site 16) and very close to the poles of the anorthosite sites 4, 5, 9, 10, and 11 in Elming et al. (2001) (Fig. 10b), which have been dated at 1760–1770 Ma. A mean pole C is calculated from the dykes and the anorthosites at Plat. = 28.8° and Plon. = 169.8° (A_{95} = 4.4°).

A mean pole D is calculated from three site means of anorthosites in Elming et al. (2001), representing a magnetization age of ca. 1740 Ma. This mean pole is not significantly different from the mean pole C and these poles can also be combined into a common mean pole (C + D, Table 4, Fig. 10c), thus representing an age of ca. 1740–1770 Ma. The mean pole C + D fulfils most of the reliability criteria of Van der Voo (1990) i.e. (1) well determined magnetization age, (2) sufficient number of samples, (3) adequate demagnetization etc., (4) positive result of field test, (5) structural control and coherence with craton, (6) reversals are present, (7) no resemblance to younger palaeopoles, which gives a quality index Q = 7. Also the criteria for a key pole are fulfilled (Buchan et al., 2000).

The palaeomagnetic poles calculated from the dykes representing the youngest magnetization ages are the poles of the Khmelnik dyke (site 3) dated at ca. 1720 Ma and poles 5 and 6b. The chemical composition of the dykes of sites 5 and 6 correspond to the Osnitsk post-granite group of dykes, while the pole positions are close to that of the anorthosite related Khmelnik dyke. The low T_{ub} component 6b, possibly carried by pyrrhotite, and the magnetization of the dyke at site 5 is thus probably the result of remagnetization and younger than the high T_{ub} component (6a), which has an interpreted age of ca 2.0 Ga. Poles 3, 5 and 6b have been combined with poles calculated from the dyke (site 14) and the ca 1720 Ma anorthosite, site 13, in Elming et al. (2001) and they form a ca. 1720 Ma mean pole position (E in Fig. 10c) at Plat. = 10.7°, Plon. = 163.2° (A_{95} = 10.2°).

The site mean pole positions calculated for the basic dykes and the poles in Elming et al. (2001) plot from low latitudes at ca. 180°E longitude (or 0°E depending on polarity; sites 12 and 15) to high northern latitudes down to the equator at 170–180°E longitude (Fig. 10a and b). The gradual change in pole positions is reflected by a gradual change in age from ca. 2060 to 1720 Ma, which means this distribution of poles may be explained by apparent polar wander.

Geological and geophysical data indicate several stages of accretion and westward growth of the Palaeoproterozoic crust, with magmatism related to eastward subduction of oceanic crust beneath Sarmatia at ca. 2100 Ma (Claesson et al., 2001) and formation of juvenile Palaeoproterozoic crust.

When plotting the mean of site poles together with key-poles from the Fennoscandian Shield (Table 5) there is a significant missmatch (Fig. 10c). This suggests that the Ukrainian Shield (and Sarmatia) was not in its present position relative to Fennoscandia at times of ca. 1720 Ma and earlier.

Rotating the 1770–1760 Ma mean pole C for Ukraine to a position close to the 1770 Ma key pole for Fennoscandia (Rotation parameters: Lat. = 48°N, Lon. = 42.2°E, Rot. Angle = 43°; Table 4; Fig. 10d) the trend of apparent polar wander for Ukraine expressed by the rotated mean poles C, D (and C + D) and E coincide with the trend of apparent polar wander of Fennoscandia for the period 1770–1630 Ma. For the comparison of poles older than ca. 1800 Ma, there are no Ukrainian key poles, but the poles are

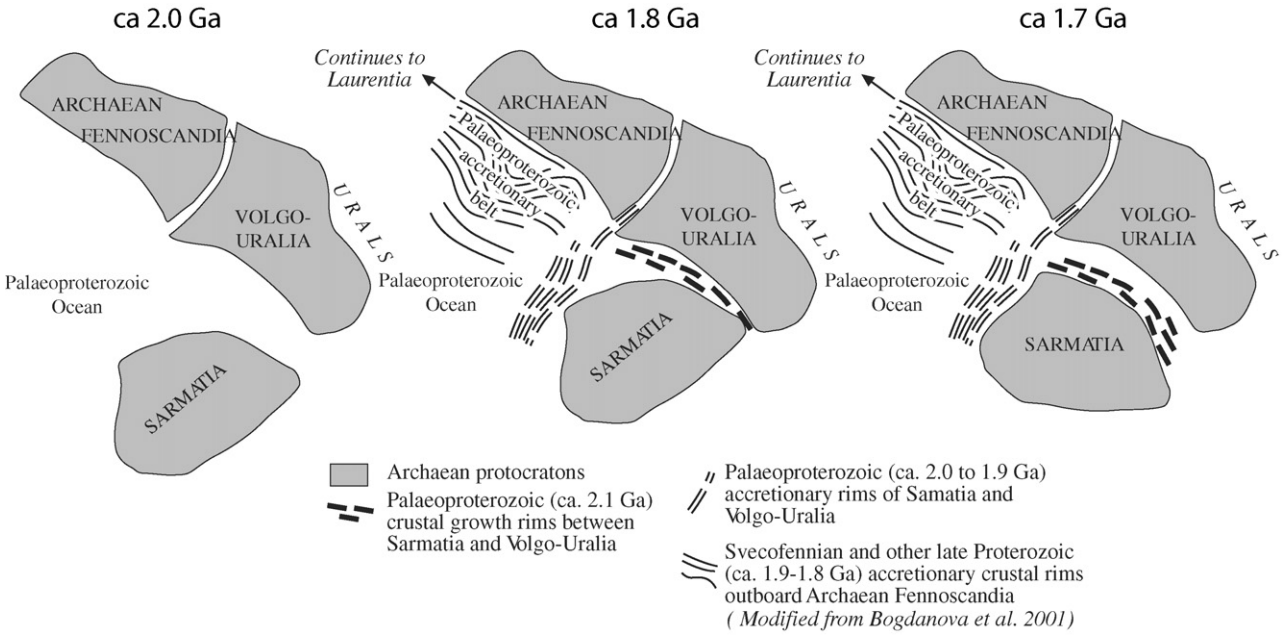


Fig. 11. The accretion of the Ukrainian Shield (and Sarmatia) to Fennoscandia based on the results of this study. The time for accretion and rotation of Sarmatia is estimated at ca 1900–1800 Ma and 1720–1660 Ma, respectively.

probably reliable. For these times the poles of the Ukraine and Fennoscandia are not coeval, which makes a direct tectonic comparison impossible. However, for poles older than ca. 1900 Ma the distributions of poles are very different, which suggests a different tectonic history for the shields. In our interpretation, the Ukrainian Shield (and Sarmatia) accreted to Fennoscandia at ca. 1900–1800 Ma and then rotated into its present position relative to Fennoscandia.

The crustal terranes adjacent to the north-western part of Sarmatia are bounded by fault zones and $^{40}Ar/^{39}Ar$ age data (Bogdanova et al., 1996) suggest a tectonomagmatic peak in one of these fault zones at 1706–1660 Ma. Bogdanova et al. (2006) suggested that post collision extensional tectonics was of crucial importance from ca. 1800 to 1740 Ga and even later at ca. 1710–1670 Ma. This may be related with the rotation of the Ukrainian Shield into its final position relative to Fennoscandia (Fig. 11).

Based on geological and geochronological similarities, the existence of a Palaeo-Mesoproterozoic supercontinent, Columbia, has been suggested (e.g. Hoffman, 1989). The hypothesis is partly supported by palaeomagnetic data suggesting an assembly of Fennoscandia, Laurentia and Amazonia into such a supercontinent at ca. 1800 Ma (Buchan et al., 2000; Pesonen et al., 2003; Bispo-Santos et al., 2008), i.e. the time at which we suggest the Ukrainian Shield accreted to Fennoscandia. In some reconstructions the pre-Rodinia basement of West Africa was in the late Palaeoproterozoic attached to the south-western part of Sarmatia (e.g. Hou et al., 2008), while others suggest the North China craton was in a position between Amazonia and Sarmatia (Kusky et al., 2007; Bispo-Santos et al., 2008). Some authors (e.g. Rogers and Santos, 2002; Zhao et al., 2004) have suggested a short life time for Columbia with a breakup that started at ca. 1.6 Ga. It may be possible that the rotation of the Ukrainian Shield (and Sarmatia) relative Fennoscandia is related to a breakup of the supercontinent at 1700–1600 Ma, which ended with the separation of Laurentia from Fennoscandia at ca. 1260 Ma (e.g. Elming and Mattsson, 2001; Pesonen et al., 2003).

Oblique collision of Sarmatia to Fennoscandia and rotation in the late phase of accretion may be another possibility. Oblique collision was suggested for the accretion of Volgo-Uralia and the Archaen part of Fennoscandia (Bogdanova et al., 1996).

7. Conclusion

With this study of basic dykes, the palaeomagnetic database for the Ukrainian Shield is extended and new reliable poles are determined. Three generations of basic dykes are indicated from the palaeomagnetic data and the results of the U–Pb and $^{40}Ar/^{39}Ar$ age determinations of this study together with other age data support an interpretation of the calculated pole positions to reflect an apparent polar wander between ca. 2060 and 1720 Ma. This age distribution of poles is also supported by the geochemical signatures of the dykes, which are generally different for the three palaeomagnetically grouped dykes.

The misfit in apparent polar wander between the Ukrainian and Fennoscandian shields indicates that the present position of the Ukranian Shield relative Fennoscandia is not the same as for ca. 1770 Ma ago. On the basis of palaeomagnetic, age, geochemical and geological data a scenario is presented here in which the Ukrainian Shield (and Sarmatia) collided with Fennoscandia at ca.1900–1800 Ma and after that, at ca. 1720–1660 Ma, the Ukrainian shield rotated ca. 43° into its present position.

Acknowledgement

This work was supported financially by the Swedish Research Council and the Swedish Royal Academy of Science. The authors wish to thank Richard Ernst and an anonymous reviewer for their comments, which significantly improved the manuscript. Thanks are also due to Kjell Billström at the Swedish Museum of Natural History for his efficient guidance and help with the U–Pb age determinations and to Milan Vnuk for drafting the figures. Nordsim is operated under and agreement between the Swedish Museum of Natural History, the Geological Survey of Finland and the research funding agencies of Denmark, Norway and Sweden. This is Nordsim publication #252.

Appendix A.

Sam. name		#Mg	SiO_2	TiO_2	Al_2O_3	Fe2O3	MnO
Bekhi	Bekhi dyke	26.5	51.48	2.93	12.96	15.91	0.17
Average	Tomashgorod dyke	45.8	49.93	1.26	15.26	13.97	0.19
U98-49	Susly old dyke	70.0	48.81	0.87	10.61	10.39	0.18
U98-57	Susly old dyke central part	73.4	49.74	0.74	10.21	9.50	0.17
05-C3	Susly old dyke central part	69.3	49.42	0.62	9.35	10.89	0.18
Average		70.9	49.32	0.74	10.06	10.26	0.18
U98-75	Susly young dyke	41.9	50.97	1.68	14.14	14.20	0.21
U98-78	Susly young dyke margin	43.8	50.52	1.62	14.46	13.61	0.21
05-C4	Susly young dyke	41.5	50.93	1.64	13.61	14.27	0.21
Average		42.4	50.81	1.65	14.07	14.03	0.21
03-T18	Small dyke in Tomashgorod quarry	38.8	52.32	1.92	14.61	13.68	0.19
U98-27	Tomashgorod, dyke 6 central aprt	60.2	50.05	0.75	15.64	10.12	0.17
U98-35	Tomashgorod, dyke 4 central aprt	48.4	63.34	0.604	15.78	5.9	0.09
U96-157	Tomashgorod, dyke 4 margin	50.4	51.63	1.00	17.23	10.11	0.15
06-E19-K	Dyke in Tomashgorod quarry	36.3	55.22	1.74	13.76	13.48	0.20
U157-97	Yasnagorka, dyke	68.8	46.31	0.75	12.77	11.80	0.18
06-E30	Khmelnik dyke	42.8	47.96	2.54	15.88	15.13	0.18
U55-97	Khmelnik dyke	41.2	48.78	2.59	15.51	15.37	0.18
Average (including data by Tsymbal)		39.7	48.49	2.76	14.49	15.90	0.18
U2-97	Bazavluk dyke	50.6	51.51	0.79	14.38	12.29	0.20
U96-180	Smela dyke	16.8	59.50	1.45	14.83	10.71	0.12

References

Ahmetshina, A.K., 1975. Analysis of the regularities of orientation of the dykes on the Ukrainian and Canadian Shields. Reports of Ac. of Sci. of the Ukraine, Cep. B, 1, 3–5.

Amelin, Y.V., Heaman, L.M., Verchogliad, V.M., Skobelev, V.M., 1994. Geochronological constraints on the emplacement history of an anorthosite-rapakivi granite suite: U–Pb zircon and baddeleyite study of the Korosten complex, Ukraine. Contrib. Mineral. and Petrol. 116, 411–419.

Bialowoska, A., Bakun-Czubarow, N., Fedoryshyn, Y., 2002. Neoproterozoic flood basalts of the beds of the Volhynian Series (East European Craton). Geol. Q. 46, 37–57.

Bispo-Santos, F., DÁgrella-Filho, M.S., Pacca, I.G., Janikian, L., Trinidade, R.I.F., Elming, S.Å., Silva, J.A., Barros, M.A.S., Pinho, F.E.C., 2008. Columbia revisited: Paleomagnetic results from the 1790 Ma colider volcanics (SW Amazonian Craton Brazil). Precambrian Res. 164, 40–49.

Bleeker, W., 2003. The late Archean record: a puzzle in ca. 35 pieces. Lithos 71, 99–134.

Bogdanova, S.V., 1993. The three-segment hypothesis for the East European Craton. Terra Nova Abstr. 5, 313.

Bogdanova, S.V., Pashkevich, I.K., Gorbatschev, R., Orlyuk, M.I., 1996. Riphean rifting and major Palaeoproterozoic crustal boundaries in the basement of the East European Craton: geology and geophysics. Tectonophys 268, 1–21.

Bogdanova, S., Gorbatschev, R., Grad, M., Janik, T., Guterch, A., Kozlovskaya, E., Motuza, G., Skridlaite, G., Starostenko, V., Taran, L., EUROBRIDGE, POLONAISE Working Groups, 2006. EUROBRIDGE: new insight into the geodynamic evolution of the East European Craton. In: Gee, D.G., Stephensson, R.A. (Eds.), European Lithosphere Dynamics, vol. 32. Geological Society, London, Memoirs, pp. 599–625.

Buchan, K., Mertanen, S., park, R.G., Pesonen, L.J., Elming, S.-Å., Abrahamsen, N., Bylund, G., 2000. Comparing the drift of Laurentia and Baltica in the Proterozoic. The importance of key palaeomagnetic poles. Tectonophysics 319, 167–198.

Bogdanova, S.V., Page, L.M., Skridlaite, G., Taran, L.N., 2001. Proterozoic tectonothermal history in the western part of the east European Craton: $^{40}Ar/^{39}Ar$ geochronological constraints. Tectonophys 339, 39–66.

Claesson, S., Bogdanova, S.V., Bibikova, E.V., Gorbatschev, R., 2001. Isotopic evidence for Palaeoproterozoic accretion in the basement of the East European Craton. Tectonophys 339, 1–18.

Claesson, S., Bibikova, E., Bogdanova, S., Skobelev, V., 2006. Archaen terranes, Palaeoproterozoic reworking and accretion in the Ukrainian Shield, East European Craton. In: Gee, D.G., Stephensson, R.A. (Eds.), Europeran Lithosphere Dynamics, vol. 32. Geological Society, London, Memoirs, pp. 645–654.

Claesson, S., Bibikova, E., Bogdanova, S., Shumlysnskyy, L., Skobelev, V., 2008. Palaeoproterozoic evolution of the northwestern Ukrainian Shield. In: 33rd International Geological Congress, Oslo (Abstract HPP-04).

Compston, W., Sambridge, M.S., Reinefrank, R.F., Moczydlowska, M., Vidal, G., Claesson, S., 1995. Numerical ages, volcanic rocks and the earliest faunal zone within the Late Precambrian of east Poland. Journal of the Geological Society, London 152, 599–611.

Davis, D.W., Blackburn, C.E., Krogh, T.E., 1982. Zircon U–Pb ages from the Wabigoon-Manitou Lake region, Wabigoon Subprovince, northwest Ontario. Canadian Journal of Earth Sciences 19, 254–266.

Elming, S.-Å., Pesonen, L.J., Leino, M.A.H., Khramov, A.N., Mikhailova, N.P., Krasnova, A.F., Mertanen, S., Bylund, G., Terho, M., 1993. The drift of the Fennoscandian and Ukrainian shields during the Precambrian: A palaeomagnetic analysis. Tectonophys 223, 177–198.

Elming, S.-Å., Mikhailova, N.P., Kravchenko, S., 2001. Palaeomagnetism of Proterizoic rocks from the Ukrainian shield and a tectonic reconstruction versus Fennoscandia. Tectonophys 339, 19–38.

Elming, S.-Å., Mattsson, H., 2001. Post Jotnian basic intrusions in the Fennoscandian Shield, and the break up of Baltica from Laurentia: a palaeomagnetic and AMS study. Precambrian Res. 108, 215–236.

Elming, S.Å., Kravchenko, S., 2005. A palaeomagnetic study of Mesoproterozoic basic dykes from the Ukrainian Shield and the amalgamation of the shield with Fennoscandia. In: Fifth international dyke conference: Dyke swarms–time markers of crustal evolution, J. Vuollo and S. Mertanen (Eds). Rovaniemi 2005, abstract, p. 12.

Elming, S.-Å., Kravchenko, S.N., Layer, P., Rusakov, O.M., Glevasskaya, A.M., Mikhailova, N.P., Bachtadse, V., 2007. Palaeomagnetism and $^{40}Ar/^{39}Ar$ age determinations of Ediacaran traps from the southwestern margin of the East European Craton, Ukraine: relevance to the Rodinia break-up. Journal of the Geological Society, London 164, 1–14.

Gorbatschev, R., Bogdanova, S.V., 1993. Frontiers in the Baltic Shield. Precambrian Res. 64, 3–22.

Hoffman, F.P., 1989. Speculations on Laurentia′s first gigayear (2.0–1.0 Ga). Geology 17, 135–138.

Hou, G., Santosh, M., Qian, X., Lister, G.S., Li, J., 2008. Configuration of the Late Paleoproterozoic supercontinent Columbia: Insights from radiating dyke swarms. Gondwana Res. 14, 395–409.

Jaffey, A.H., Flynn, K.F., Glendenin, L.E., Bentley, W.C., Essling, A.M., 1971. Precision measurements of half-lives and specific activities of ^{235}U and ^{238}U. Phys. Rev., C4 5, 1889–1906.

Kirschvink, J.L., 1980. The least-square line and plane and the analysis of paleomagnetic data. Geophys. J. R. Astronom. Soc. 62, 699–718.

Krogh, T.E., 1973. A low contamination method for hydrothermal decomposition of zircon and extraction of U and Pb for isotopic age determinations. Geochemica et Cosmochemica Acta 37, 485–494.

Krogh, T.E., 1982. Improved accuracy of U–Pb zircon ages by creation of more concordant systems using an air abrasion technique. Geochemica et Cosmochemica Acta 46, 637–649.

Krutihovskaja, Z.A., Pashkevich, I.K., Podolanko, S.M., 1976. The regularities of distribution of the dyke complexes of the Ukrainian Shield. Geophys. Coll. Act. Sci. Ukraine 74, 61–64 (in Russian).

Kusky, T., Li, J., Santosh, M., 2007. The paleoproterozoic North Hebei Orogen: North China craton′s collosional suture with Columbia supercontinent. Gondwana Res. 12, 4–28.

Lanphere, M.A., Dalrymple, G.B., 2000, First-principles calibration of ^{38}Ar tracers: Implications for the ages of $^{40}Ar/^{39}Ar$ fluence monitors, U.S. Geological Survey Professional Paper 1621, 10 pp.

Layer, P.W., Hall, C.M., York, D., 1987. The derivation of $^{40}Ar/^{39}Ar$ age spectra of single grains of hornblende and biotite by laser step heating. Geophys. Res. Lett. 14, 757–760.

Layer, P.W., 2000. $^{40}Ar/^{39}Ar$ age of the El'gygytgyn impact event, Chukotka, Russia. Meteor. Planet. Sci. 35, 591–600.

Le Maitre, R., Bateman, P., Dudek, A., Keller, J., Lameyre, J., La Bas, M., Sabine, P., Sorensen, H., Streckeisen, A., Woolley, A., Zanetti, B., 1989. A classification of igneous rocks and glossary of terms: Recommendations of the International Union of Geological Sciences Subcommision on the Systematics of igneous rocks, In: R. Le Maitre (Ed.). Blackwell, Oxford, 193 pp.

Ludwig, K.R., 1999. Isoplot/Ex Version 2.03. A Geochronologocal Toolkit for Microsoft Excel. Berleley Geochronology Center, Special Publication 1, 43 p.

Ludwig, K.R., 2003. A Geochronological Toolkit for Microsoft Excel, Berkeley Geochronological Center (2003), Special Publication, No. 4, p. 75.

McDougall, I., Harrison, T.M., 1999. Geochronology and Thermochronology by the $^{40}Ar/^{39}Ar$ method, second ed. Oxford University Press, New York, 269 pp.

Meert, J.G., 2002. Paleomagnetic evidence for a Paleo-Mesoproterozoic supercontinent Columbia. Gondwana Res. 5, 207–216.

Mertanen, S., Pesonen, L.J., 1994. Preliminary results of a palaeomagnetic and rock magnetic study of the Proterozoic Tsoumasvarri intrusion, northern Fennoscandia. Precambrian Res. 69, 25–50.

Mertanen, S., Halls, H.C., Vuollo, J.I., Pesonen, L.J., Stepanov, V.S., 1999. Palaeomagnetism of 2.44 Ga mafic dykes in Russian Karelia, eastern Fennoscandian Shield–implications for continental reconstructions. Precambrian Res. 98, 197–221.

Mikhailova, N.P., Karsonova, A.J., Kotlovskaya, F.L., Skatalov, N.N., 1985. Correlation scheme of dyke formations from the Ukrainian Shield with elements of palaeomagnetic and radioisotopic data. Geol. J. 45 (6), 38–45.

Neuvonen, K.J., Korsman, K., Kouvo, O., Paavola, J., 1981. Palaeomagnetism and age relations of the rocks in the main Sulphide Ore Belt in central Finland. Bull. Geol. Soc. Finl. 53, 109–133.

Pesonen, L.J., Elming, S.-Å., Mertanen, S., Pisarevski, S., DĭAgrella-Filho, M.S., Meert, J., Schmidt, P.W., Abrahamsen, N., Bylund, G., 2003. Paleomagnetic configuration of continents during the Proterozoic. Tectonophys 375, 289–324.

Pisarevsky, S.A., Sokolov, S.J., 2001. The magnetostratigraphy and a 1780 Ma palaeomagnetic pole from the red sandstones of the Vazhinka River section, Karelia, Russia. Geophys. J. Int. 146, 531–538.

Rogers, J.J.W., 1996. A history of the continents in the past three bilion years. J. Geol. 104, 91–107.

Rogers, J.J.W., Santos, M., 2002. Configuration of Columbia, a Mesoproterozoic supercontinent. Gondwana Res. 5, 5–22.

Samson, S.D., Alexander, E.C., 1987. Calibration of the interlaboratory $^{40}Ar–^{39}Ar$ dating standard, MMhb-1. Chem. Geol. 66, 27–34.

Scherbak, D.P., Ponomarenko, A.N., Makarenko, I.D., 1995. Geochronology of the Granites from the Ingulo-Ingulets Block of the Ukrainian Shield. Geochemistry and Ore Origin, vol 21. Naukova Dumka, Kiev, pp. 74–88 (in Russian).

Scherbak N.P., Artemenko G.V., Lesnaya I.M., Ponomarenko A.N., Shumlyanskyy L.V., 2008. Geochronology of the early Precambrian of the Ukranian shield. Proterozoic. Naukova Dumka, Kiev, 2008, 240 pp (in Russian).

Schipansky, A.A., Bogdanova, S.V., 1996. The Sarmatian crustal segment: Precambrian correlations between the Voronezh Massif and the Ukrainian Shield across the Dniepr-Donets Aulacogen. Tectonophys 268, 109–126.

Shumlyanskyy, L.V., Belousova, O.A., Elming, S.-A., 2008a. New data on isotopic age of the Palaeoroterozoic gabbro-dolerites of the North-Western region of the Ukrainian Shield. Mineral. J. (Ukraine) 30 (4), 58–69.

Shumlyanskyy, L.V., Mitrokhin, O.V., Bogdanova, S.V., Bratchuk, O.M., Yakubenko, P.F., 2008b. U–Pb on zircon isotopic age of rocks of the Korsun-Novomirgorod anorthosite-rapakivi granite pluton. Geol. Ukraine 1–2, 77–85 (in Ukrainian).

Skridlaite, G., Bogdanova, S.V., Motuza, G., 1997. A major tectonic boundary in the basement of the East European Craton in Lithuania: implications from metamorphism. Terra Nova, Supplementary Volume (EUG. 9 Abstracts), p. 132.

Söderlund, U., Johansson, L., 2002. A simple way to extract baddeleyite (ZrO_2). Geochem. Geophys. Geosyst. 3 (2), doi:10.1029/2001GC000212.

Stacey, J.S., Kramers, J.D., 1975. Approximation of Terrestrial lead isotope evolution by a two-stage model. Earth Planet. Sci. Lett. 26, 207–221.

Steiger, R.H., Jaeger, E., 1977. Subcommission on geochronology: Convention on the use of decay constants in geo and cosmochronology. Earth Planet. Sci. Lett. 36, 359–362.

Stephensson, R.A., Bogdanova, S.V., Juhlin, C., 1996. Palaeozoic rifting in a 2.0 Ga Andean-type magmatic belt in the east European Craton: Structural and rheological implications. In: Seventh International Symposium on Deep Seismic Profiling of the continents, 1996, Asilomar, California, Abstract, p. 46.

Torsvik, T.H., 1986. Interactive Analysis of Palaeomagnetic Data—IAPD User Guide. University of Bergen, pp. 1–74.

Torsvik, T.H., Briden, J.C., Smethurst, M.A., 1996. Super IAPD program. Geol. Surv. Norway.

Van der Voo, R., 1990. The reliability of paleomagnetic data. Techtonophysics 184, 1–9.

Verhogliad, V.M., 1995. Age stages of the Korosten pluton magmatism. Geokhima I Metallogenia 21, 34–47 (in Russian).

Whitehouse, M.J., Kamber, B.S., 2005. Assigning dates to thin gneissic veins in high-grade metamorphic terranes: a cautionary tale from Akilia, southwest Greenland. Journal of Petrology 46, 291–318.

Wiedenbeck, M., Allé, P., Corfu, F., Griffin, W.L., Meier, M., Oberli, F., Von Quadt, A., Roddick, J.C., Spiegel, W., 1995. Three natural zircon standards for U-Th-Pb, Lu-Hf, trace element and REE analysis. Geostandards Newsletter 19, 1–23.

Williams, H., Hoffman, P.F., Lewry, J.F., Monger, J.W.H., Rivers, T., 1991. Anatomy of North America: thematic geologic portrayals of the continent. Tectonophys 187, 117–134.

York, D., Hall, C.M., Yanase, Y., Hanes, J.A., Kenyon, W.J., 1981. $^{40}Ar/^{39}Ar$ dating of terrestrial minerals with a continuous laser. Geophys. Res. Lett. 8, 1136–1138.

Zhao, G., Cawood, P.A., Wilde, S.A., Sun, M., 2002. Review of global 2.1–1.8 Ga orogens: implications for a pre-Rodinia supercontinent. Earth Sci. Rev. 59, 125–162.

Zhao, G., Sun, M., Wilde, S.A., Li, S., 2004. A Paleo-Mesoproterozoic supercontinent: assembly, growth and breakup. Earth Sci. Rev. 67, 91–123.